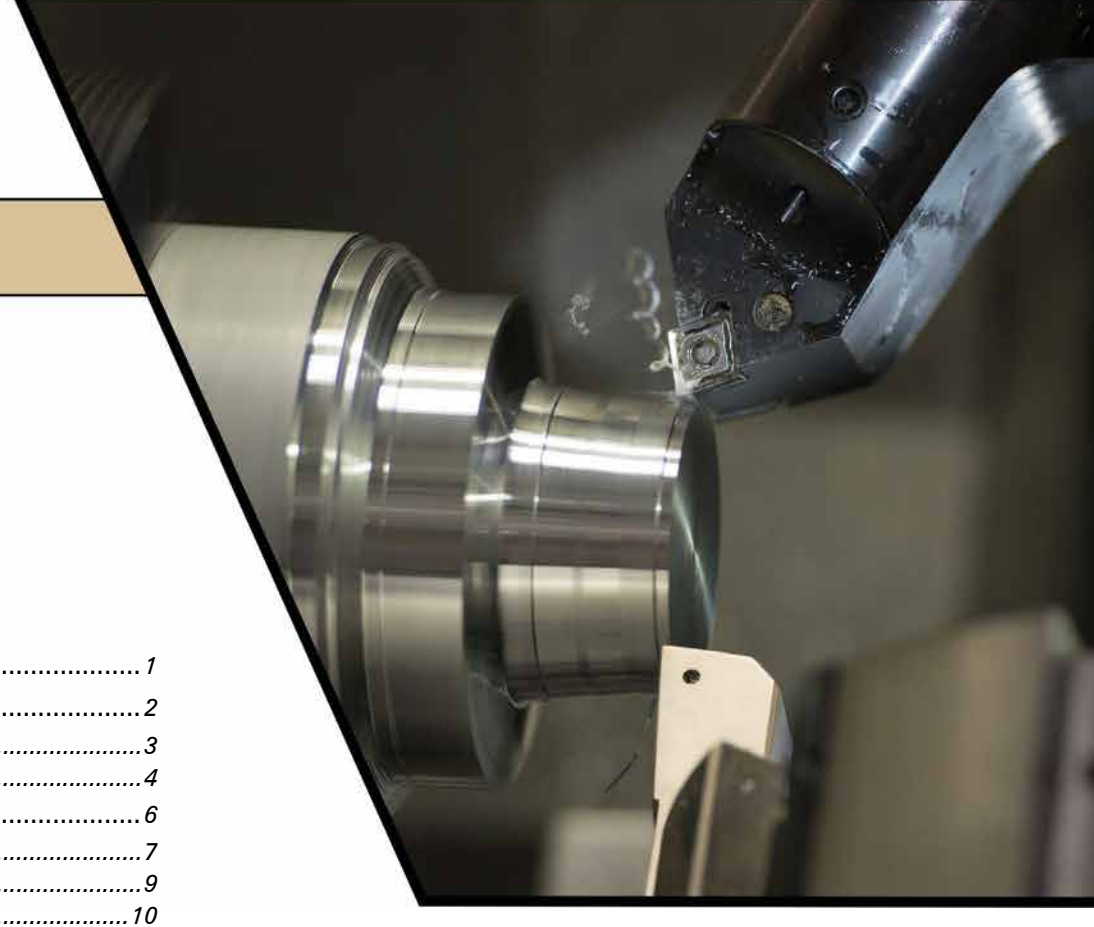
MILL-TURN APPLICATION GUIDES

Okuma P300



revision date: Dec. 2022

Contents

A: Introduction 1

B: Customizing your .machine file 2

- Default values for .machine file settings 3
- Configuring the Code Expert editor 4

C: Organizing NC output 6

- Setting the toolpath directory and stream names 7
- Configuring tool table output 9
- Selecting the work coordinate system (G15) 10
- Outputting optional stops 10
- Enabling Sub Micron mode 11
- Omitting long comments 12
- ADMAC simulation output 13
- Syncing all three streams 14
- Using the Load Monitor function 19
- Switching to IJK output for arcs 21

D: Configuring part handling operations 22

- Using the G22 torque skip function 22
- Pickoff at W0 grip position 25
- Adding a wait code to spindle commands 26
- Stopping the spindle between operations 27
- Ejecting the part 28

E: Steady rest and tailstock functions 29

- Following the tool with the steady rest 29
- Following the subspindle home with the steady rest 32
- Steady rest options (M472, M102) 35
- Supporting tailstock operations 36

Mastercam

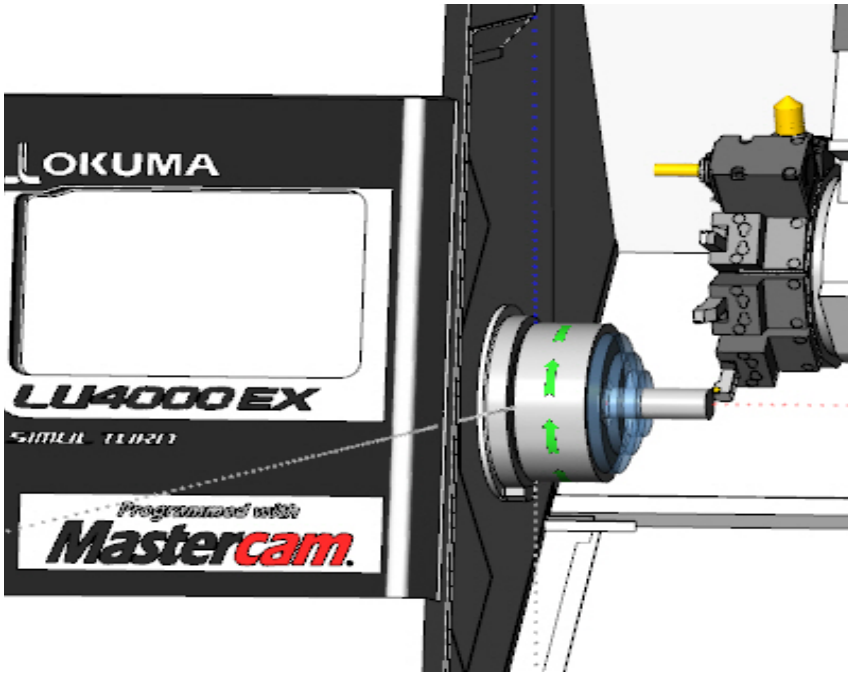
F: Approach and retract moves 38
Selecting reference positions 39
Setting the type of approach/retract motion 41
Using variables to set home positions 42
G: Configuring milling operations 44
Enabling live tooling 45
Cylindrical interpolation (G132/G133) 46
Polar interpolation (G137) and contour generation (G101/G102/G103) 48
Y-axis (G138) mode 50
C-axis clamping (M147) 51
Spindle commands for pinch/balanced milling operations 52
C-axis rotation direction 54
H: Configuring multiaxis operations 55
Pole handling 55
Feed rates for multiaxis operations 56
I: Configuring turning operations 57
Turning with both turrets on the same spindle (non-pinch turn) 57
Entering a Y offset 58
Turning on the tool spindle for lathe drill operations 59
Infeed pattern for lathe threading 60
J: Using coolant 61
Additional coolant options 61
Default coolant options 62

Copyright © 2022 Mastercam—All rights reserved.
Terms of Use—Use of this document is subject to the Mastercam End User License Agreement.

A: Introduction

Your **.machine** file drives your entire Mill-Turn experience. A Mill-Turn **.machine** file is very different from the machine and control definition files that you might be familiar with from other Mastercam products. This application guide explains the specific options and features in your **.machine** file that support your Okuma machine. It includes the following sections:

- Customizing your .machine file
- Organizing NC output
- Configuring part handling operations
- Steady rest and tailstock functions
- Approach and retract moves
- Configuring milling operations
- Configuring multiaxis operations
- Configuring turning operations
- Using coolant



Supported machines and controls—This application guide supports all Okuma mill-turn machines with the OSP-P300 control except for the Multus B-II and U series.

- LB and LB EX
- LT and LT EX
- LU and LU EX
- Genos

Screen images and code samples might be from any of these machines. Features that do not apply to all machines are noted as such.

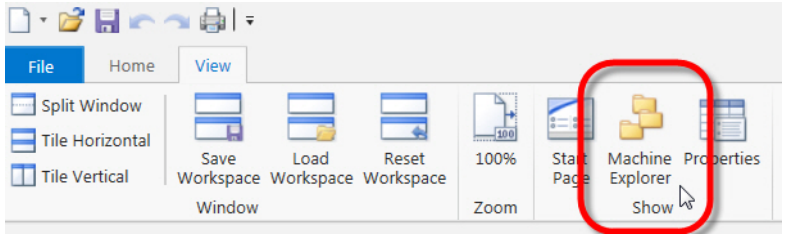
Home

Rev. date:
Dec. 2022

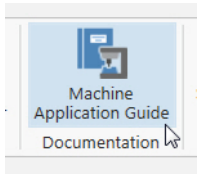
B: Customizing your .machine file

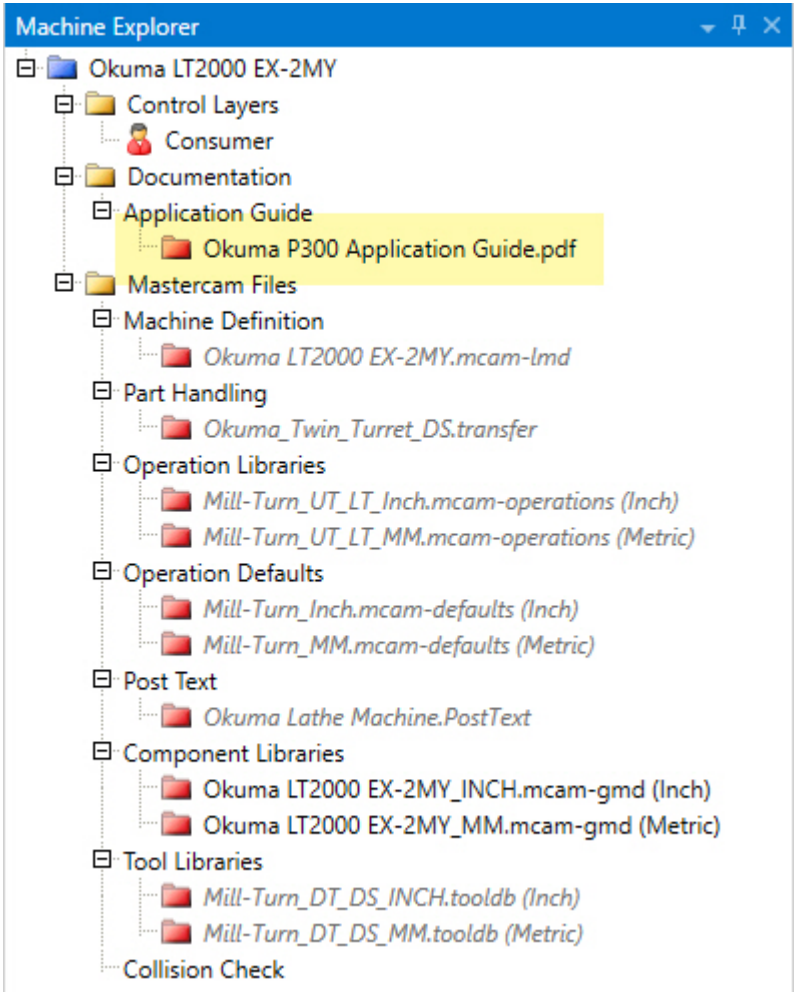
Whenever you load your **.machine** file in Mastercam to begin working on a part, Mastercam also starts Code Expert and loads your **.machine** file there as well. Code Expert is where you can make changes to your **.machine** file, such as editing default settings.

Before you can work with the **.machine** file, the Machine Explorer needs to be visible. Click the **Machine Explorer** button on the **View** tab.



You can also access this application guide directly from Code Expert: click the **Machine Application Guide** button on the **Home** tab.



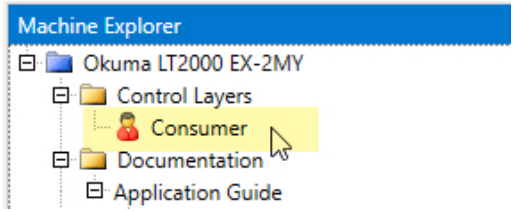


Home

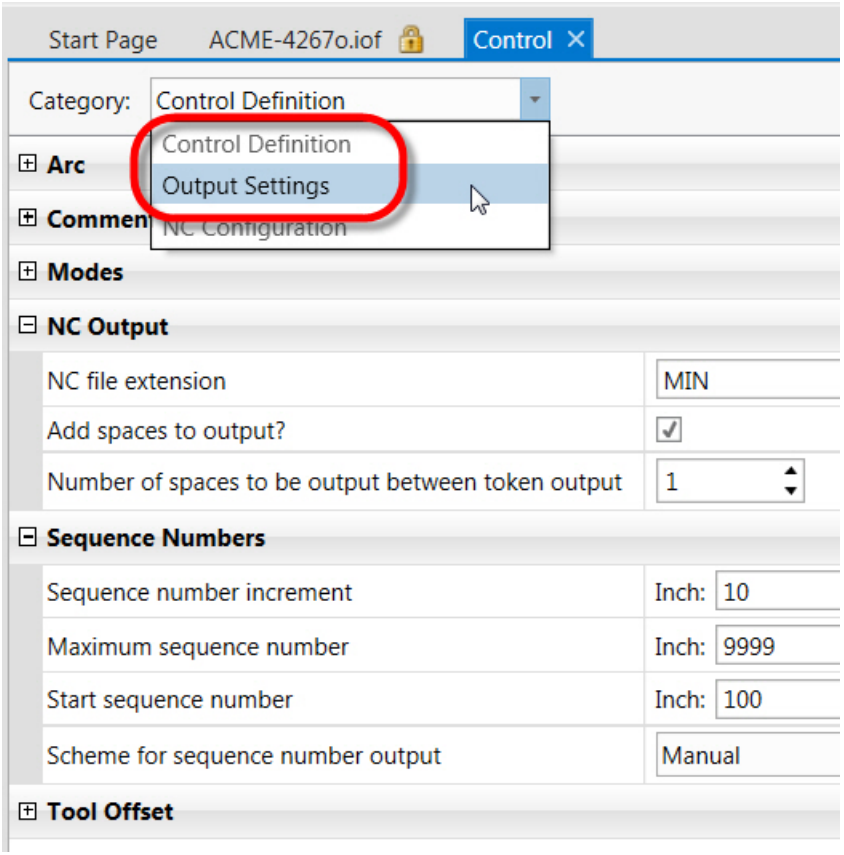
Default values for .machine file settings

Although your **.machine** file is typically supplied to you ready-to-use, it includes many settings that you can configure yourself according to your preferences and specific application needs. These include sequence and sync numbering, tool offset numbering, use of spaces in your NC file, job/shop info for your NC header, and so on.

Reach these settings by opening your **.machine** file in Code Expert and double-clicking the **Consumer** icon in the Machine Explorer.



The settings are grouped into several categories. Click the + signs to see the individual options. Some of these options (for example, sequence number settings) are very common or generic to most controls; others are specific to your individual machine.



These settings serve a wide variety of functions:

- Mimicking the control definition settings found in other Mastercam products.
- Configuring toolpath modes and cycles.
- Setting default values for Sync Manager and toolpath options.

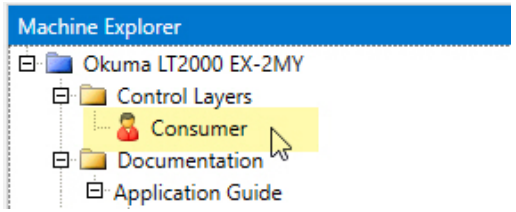
Many of these settings are self-explanatory and you can easily configure them by simply browsing the interface. The settings that are specific to this **.machine** file are described in this guide.

After making any changes, press [**Ctrl+S**] to save your **.machine** file.

Home

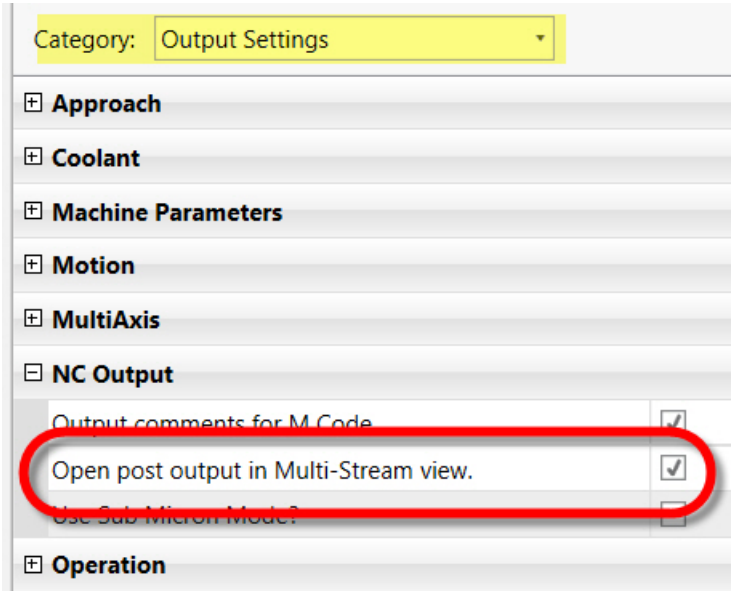
Configuring the Code Expert editor

There are several settings in the **.machine** file that you can use to configure the Code Expert editor. Open the **Consumer** layer to see these options.



Opening in multi-stream view—You can use the Code Expert editor in either single-stream or multi-stream mode. Since the P300 NC output is typically divided into two streams, you may wish to open files in multi-stream view by default.

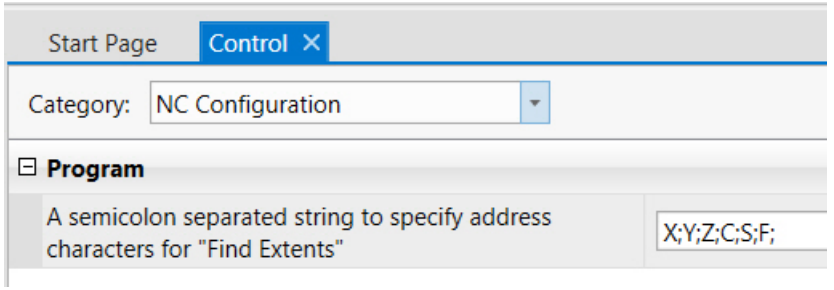
1. Go to the **Output Settings** category.
2. Open the **NC Output** group.
3. Select the **Open post output in Multi Stream view** option.

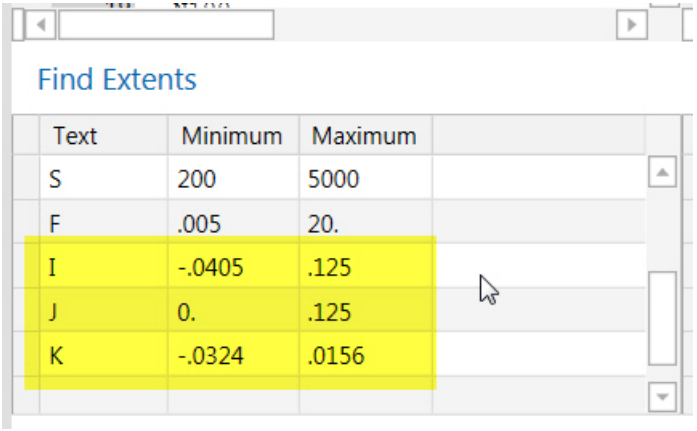


Configuring the Find Extents feature—The **Find Extents** feature in Code Expert scans your NC file and displays the minimum and maximum values for each letter address. By default, this is set to scan **X**, **Y**, **Z**, **C**, **S,** and **F**. If you wish, you can edit the set of addresses that are scanned.

1. Go to the **NC Configuration** category.
2. Open the **Program** group.
3. Enter the desired letter addresses in the list, separated by semi-colons (**;**).

For example, you can choose to add **I;J;K;** to the list. The next time you post, you will see them in the **Find Extents** table.





Home

C: Organizing NC output

Your **.machine** file includes a number of different options and default settings that let you customize how your NC output is organized and formatted.

- **Setting the toolpath directory and stream names**
- **Configuring tool table output**
- **Outputting optional stops**
- **Enabling Sub Micron mode**
- **Omitting long comments**
- **ADMAC simulation output**
- **Syncing all three streams**
- **Using the Load Monitor function**

Home

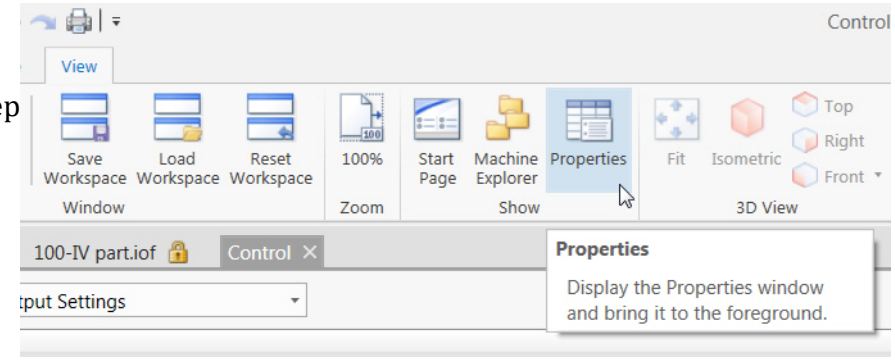
Rev. date: Dec. 2022

Setting the toolpath directory and stream names

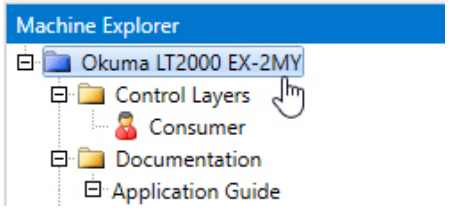
By default, your NC file is written to the following folder:

\My Mastercam 2023\Mill Turn\NC

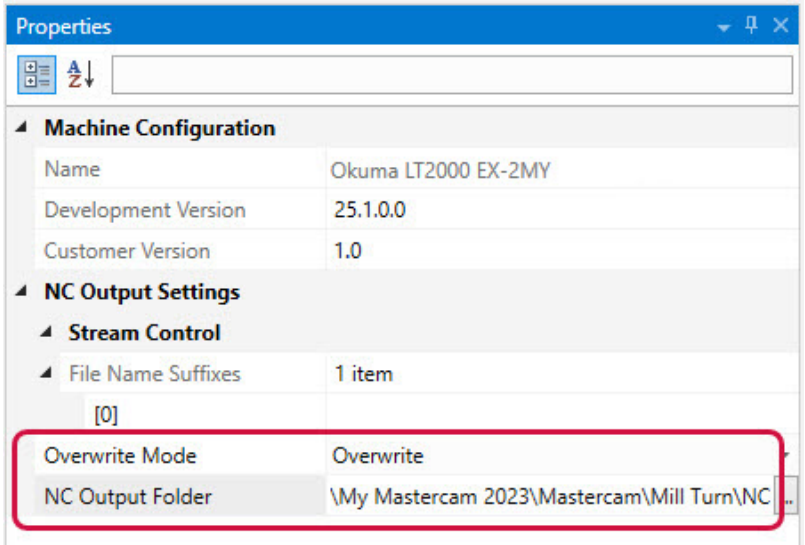
If you wish, you can set a different destination. Follow these steps:



1. Open the **.machine** file in Code Expert.
2. Make sure that the **Properties** window is displayed.



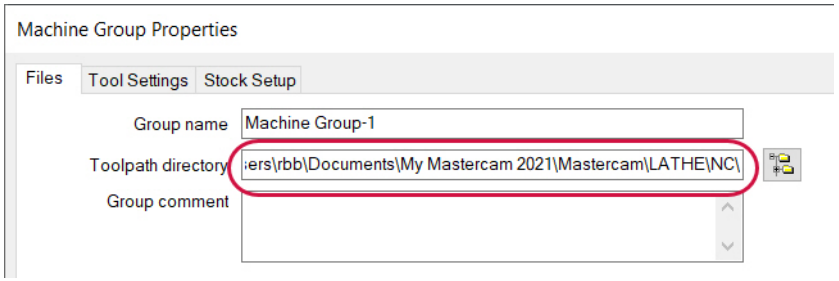
3. Click the machine name in the **Machine Explorer**.



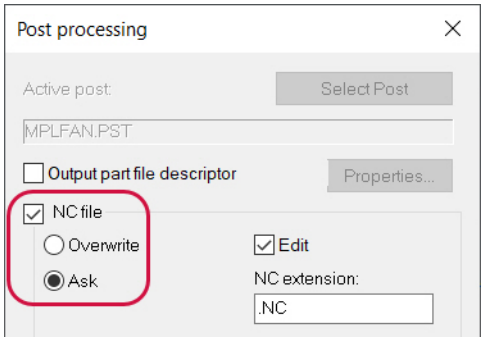
4. Select the desired **NC Output Folder**. Mastercam will write your NC files for this machine to this folder.
5. You can also choose whether Mastercam will automatically overwrite NC files with the same name, or prompt you to enter a different name. Select the desired **Overwrite Mode** to control this.
6. Press [**Ctrl+S**] to save the **.machine** file.

Home

If you are familiar with Mastercam, you are probably familiar with the **Toolpath directory** setting in the **Machine Group Properties**. This setting is not used in Mill-Turn.



Also, the **Overwrite Mode** setting replaces these settings from the **Posting** dialog box in other Mastercam products.

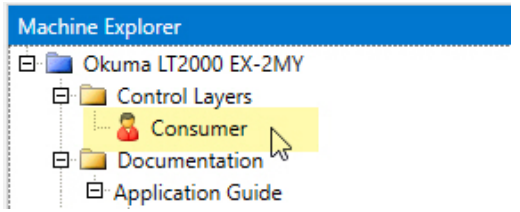


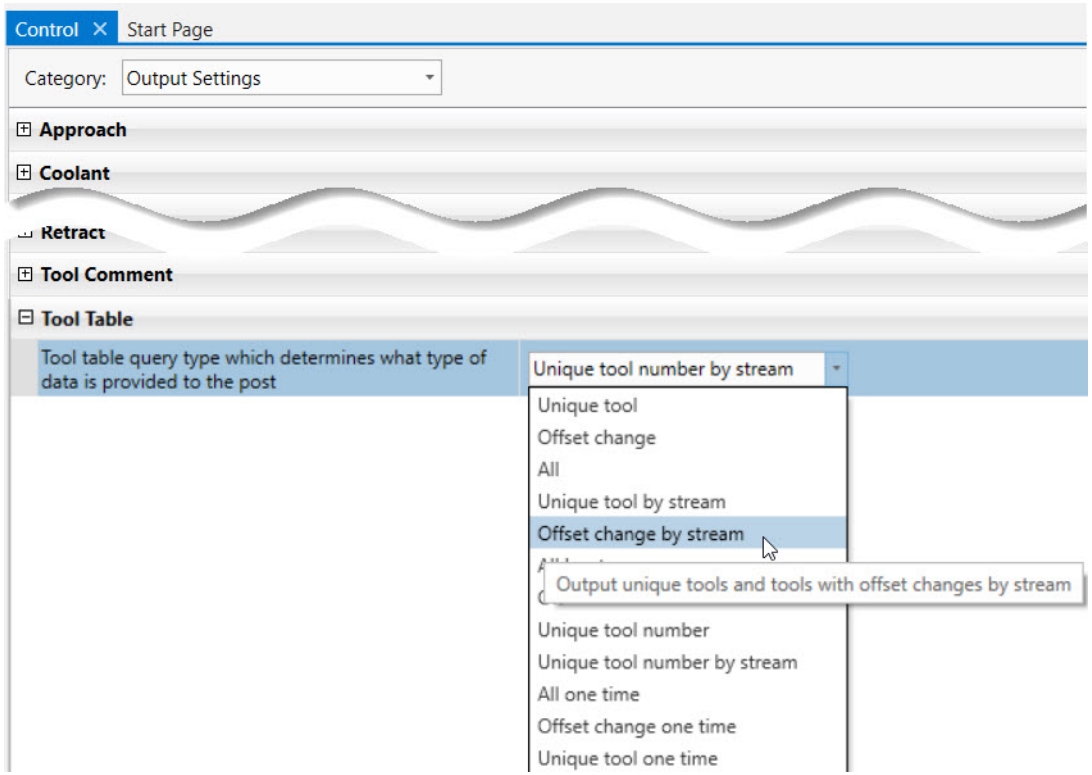
Home

Configuring tool table output

Your **.machine** file includes 12 different tool table strategies. This means that you can configure your tool table output by simply selecting the desired strategy, without any programming or edits to your post. Follow these steps:

1. Double-click the **Consumer** layer.
2. Go to the **Output Settings** category.
3. Open the **Tool table** group.
4. Select the desired strategy. Hover over each one to see a description.
5. Press [**Ctrl+S**] to save the **.machine** file.

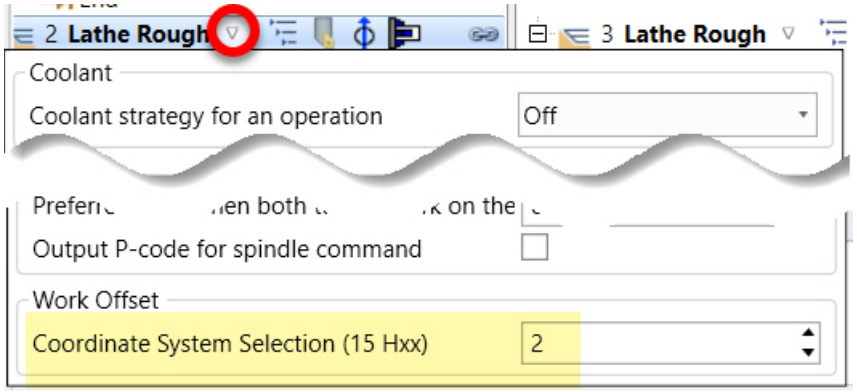




Home

Selecting the work coordinate system (G15)

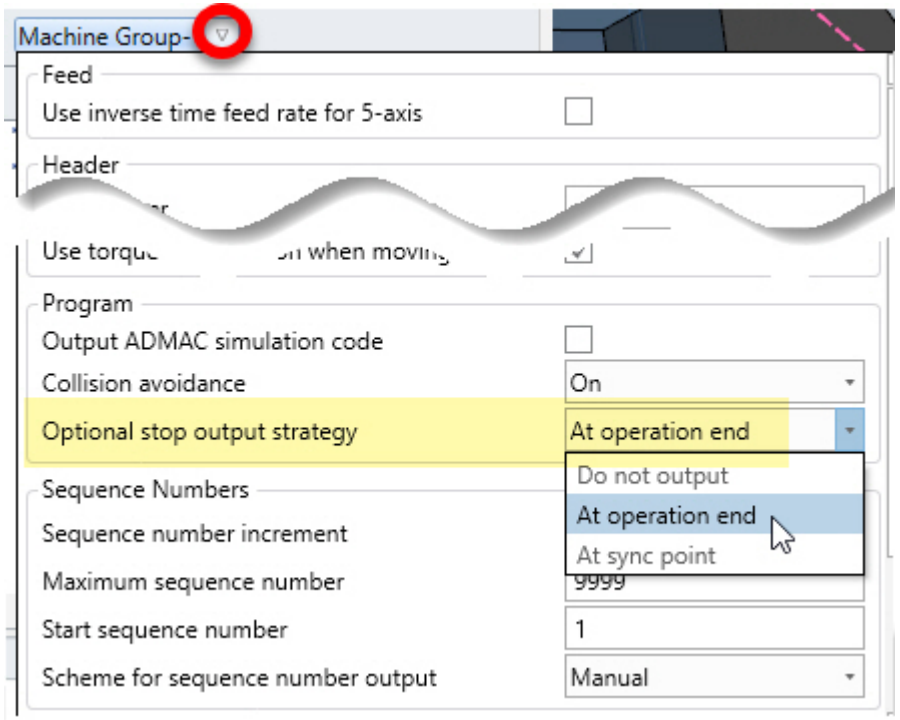
Your **.machine** file lets you select the work coordinate system/ offset for each operation. This is output before the operation on a `G15` line, in the form `G15 H__`. For each operation, enter the number as shown at right; this example would result in `G15 H2` being output in your NC file.



Outputting optional stops

Your **.machine** file includes an option that outputs optional stop codes in your NC file at selected events. You can output them at the end of each operation, or at each sync or wait code.

Click the small triangle next to the machine group name, and select the desired **Optional stop output strategy**. Then press [**Ctrl+S**] to save the setting back to your part file.

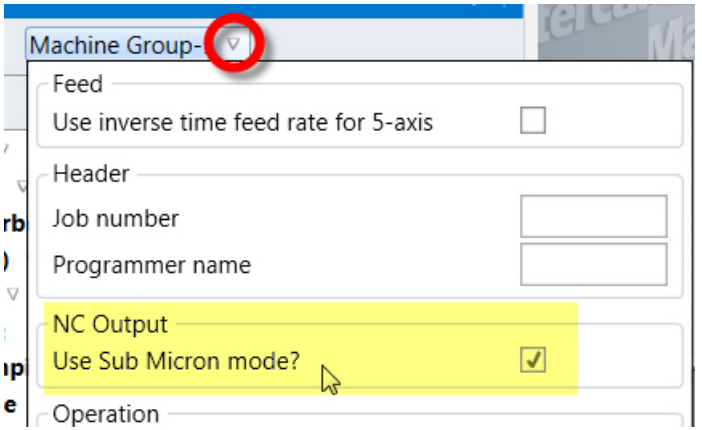


Home

Enabling Sub Micron mode

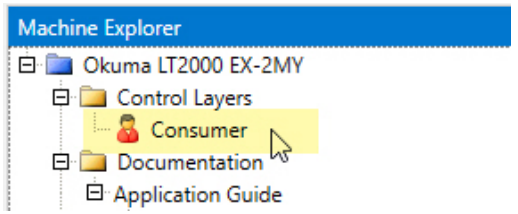
Your **.machine** file lets you select Sub Micron mode for your NC output. When working in inch mode, this results in five decimal places of precision for both linear and rotary output. In metric mode, this results in four decimal places for linear output and five decimal places for rotary output.

This setting applies to your entire part. Click the small triangle next to the machine group name and select it as shown at right.

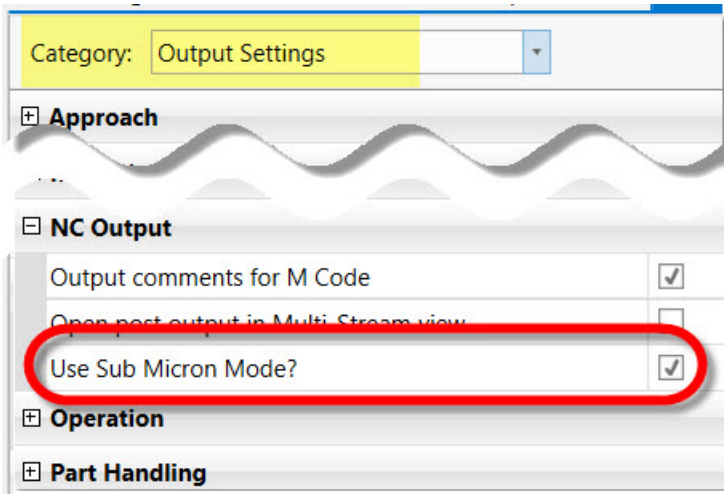


If you wish, you can enable Sub Micron mode by default. Follow these steps:

1. Open the **.machine** file in Code Expert.
2. Double-click the **Consumer** layer.



3. Select **Category: Output Settings**.
4. Go to the **NC Output** section.
5. Select the **Use Sub Micron mode** option.
6. Press [**Ctrl+S**] to save the **.machine** file.



Omitting long comments

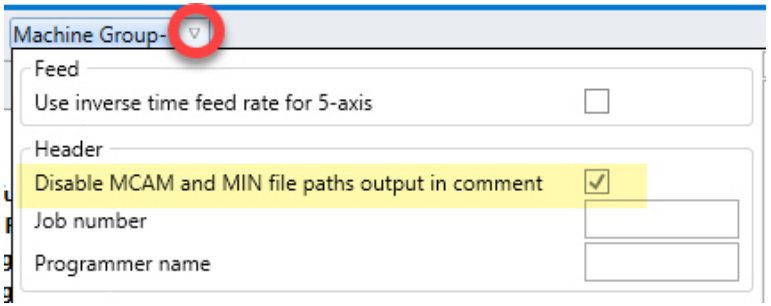
The P300 control has a limit on comment length. Often, this means that the paths used in filenames result in comments that are too long for the control, preventing the program from being loaded. The filenames and paths in this NC file are typical:

```
(START OF STREAM 1)
G13
G140
(DATE=DD-MM-YY -  08/16/2022  TIME=HH:MM -  09:01)
(MCAM FILE -  C:\Users\<user>\Documents\My Mastercam 2023\Mastercam\Parts\ACME-4465.mcam)
(NC FILE -  C:\Users\<user>\Documents\My Mastercam 2023\Mastercam\Mill Turn\NC\ACME-4465.MIN)

(TOOLS USED IN TURRET A)
(3  |   17/32 FLAT ENDMILL  | DIA. -  0.5312)
(1 |  ROUGH FACE LEFT - 80 DEG.)
(2 |  OD 55 DEG LEFT  |  DNMG-432)
(12  |   1/4 FLAT ENDMILL  | DIA. -  0.25)
(16 |  OD CUTOFF RIGHT)

SYNC=123
G140 P1
G00 W=VPVLW (MOVE SUB SPINDLE HOME)
```

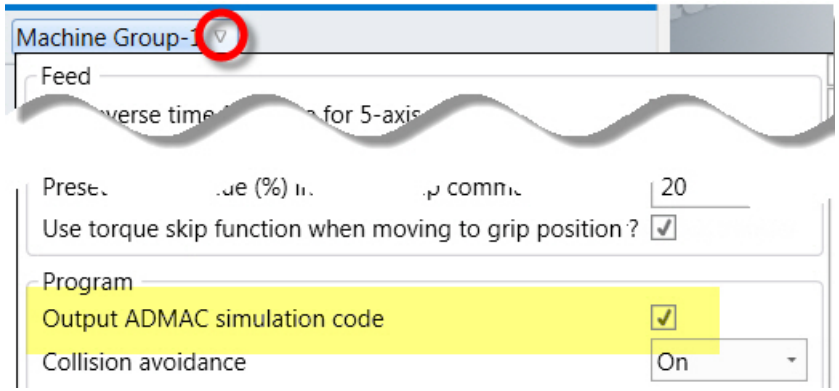
Your **.machine** file includes an option that lets you suppress these comments. Click the small triangle next to the machine group name and select the **Disable MCAM and MIN file paths…** option.



ADMAC simulation output

Your **.machine** file gives you the option to include ADMAC simulation codes in your NC file.

This setting applies to your entire part. Click the small triangle next to the machine group name and select the option as shown at right.

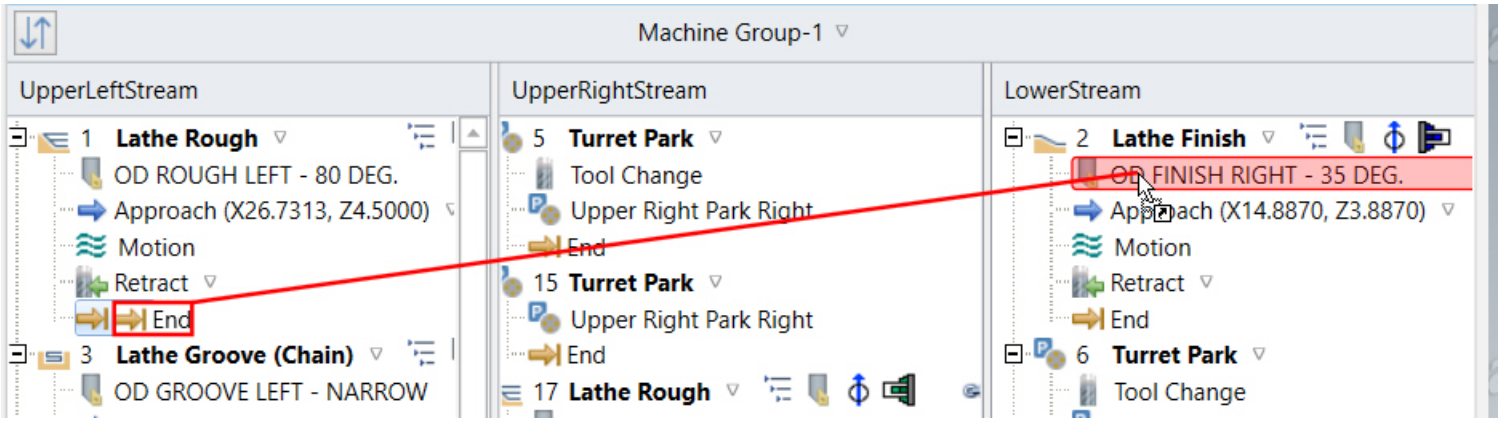


Syncing all three streams

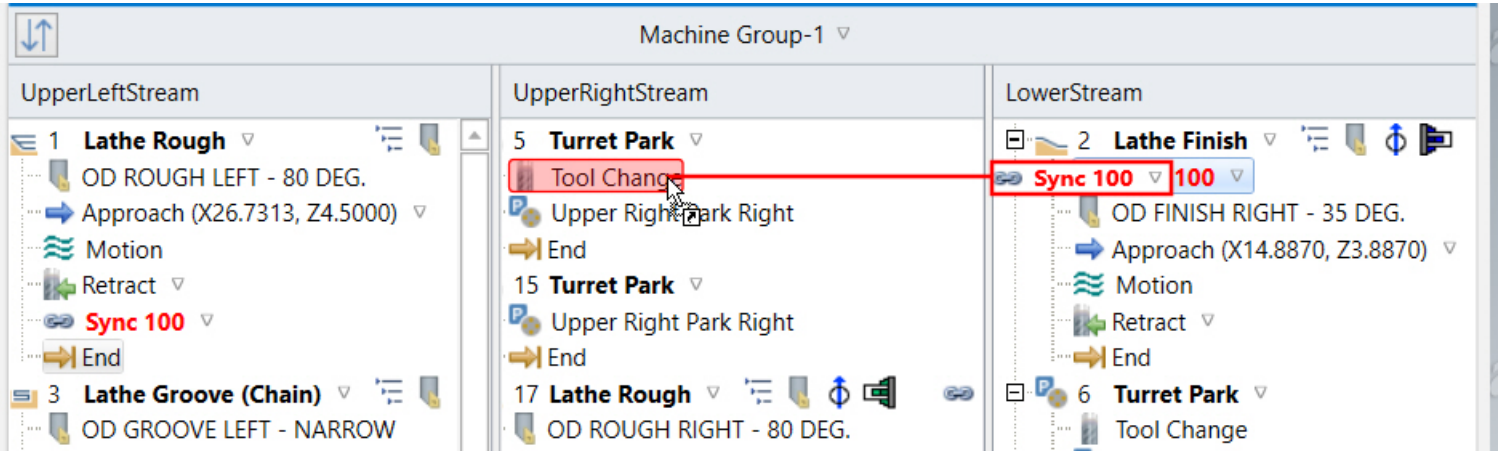
Some Okuma **.machine** files include three turrets and streams. To sync more than two streams in Mill-Turn, first create a sync between two streams. Then [**Ctrl+Click**] the sync in the second stream and drag it to the desired sync point in the third stream. The following example shows how to do this.

This example creates syncs between the first operation in each stream. Sync the end of the roughing operation (upper left turret) with the start of the finishing operation (lower turret), while parking the upper-right turret. Follow these steps:

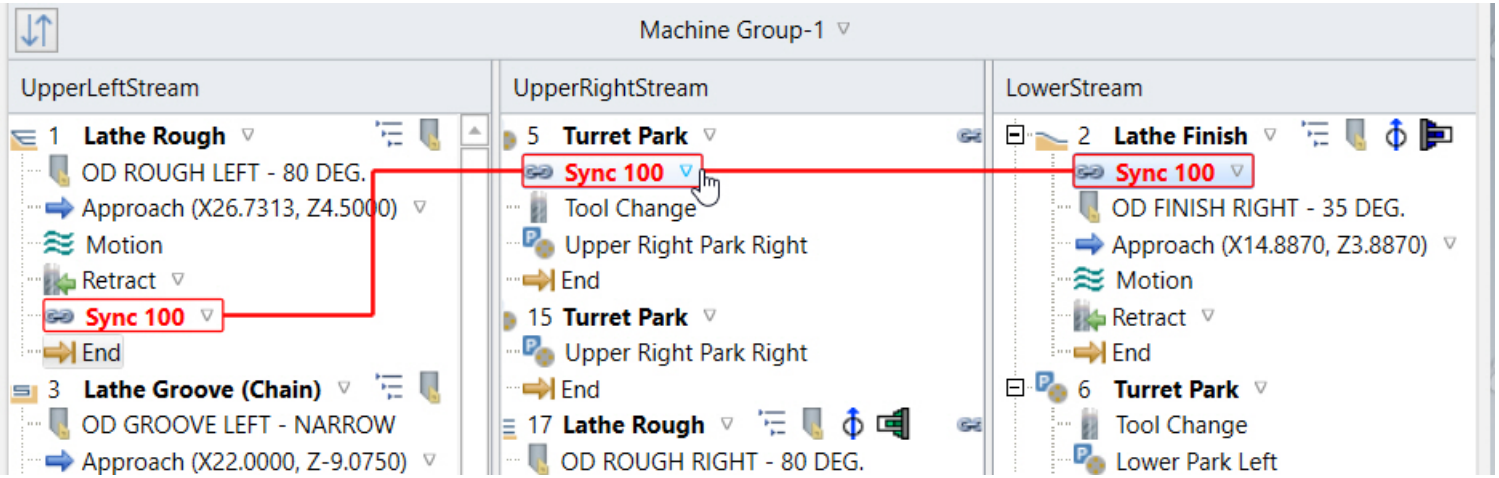
1. Sync the roughing and finishing operations. Click and drag the end of the roughing operation (stream 1) to the start of the finish operation (stream 3).



2. [**Ctrl+Click**] the sync in stream 3, then drag it to the start of the turret park operation in stream 2.

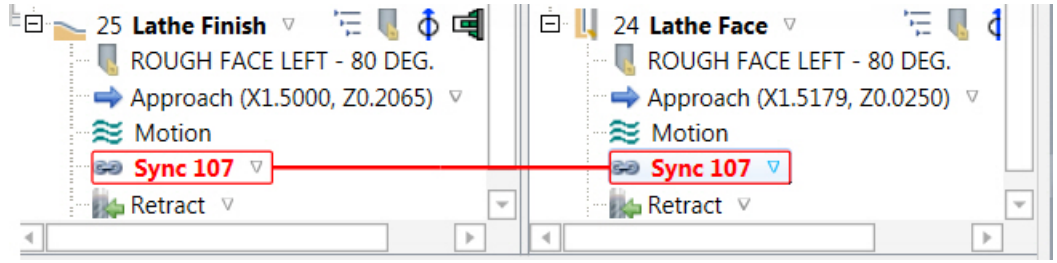


Mastercam creates the syncs as shown below:



Syncing the last operation in each stream

If your part program ends with one of the upper turrets working on the same spindle as the lower turret, create a sync between the ends of the last operation in each stream. This will ensure that one stream does not turn off the spindle while the other stream is still active.



Working with SYNC groups

Your post supports the use of the SYNC command in conjunction with the P___ wait codes. The SYNC command defines reciprocal relationships or groups among streams or turrets. By default the sync group for each P___ command includes all three turrets. However, you can customize the SYNC groups for each P___ command. This lets you fine-tune the turret synchronization across multiple syncs and streams. You can program a sync with a reciprocal relationship (in other words, turrets A and B wait for each other) or a non-reciprocal relationship (in other words, turret A waits for turret B, but turret B does not wait for turret A).

The SYNC command is output as SYNC=123, where 123 consists of any combination of the numerals 1, 2, or 3 where each number represents one of the turrets or streams:

- 1 = stream 1 (turret A / upper turret / upper left turret)
- 2 = stream 2 (turret B / lower turret / lower left turret)
- 3 = stream 3 (turret C / upper right turret)

SYNC commands and the groups they define are relative to the stream in which they are output. For example, if SYNC=23 is output in stream 1, it indicates that turret A is grouped with turrets B and C; however, if it is output in stream 2 or 3, it indicates that only turrets B and C are grouped together, without turret A.

Including the local turret in the SYNC command is not required, nor is the order of the stream numbers significant. This means that,

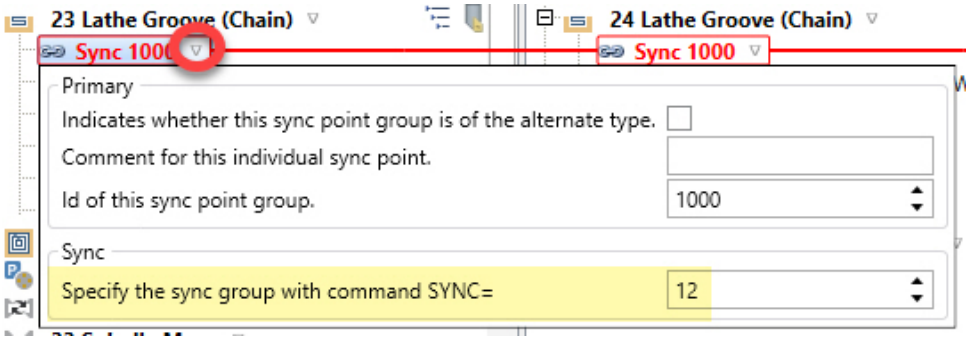
Home

assuming they are each output in stream 2, the following SYNC commands are equivalent:

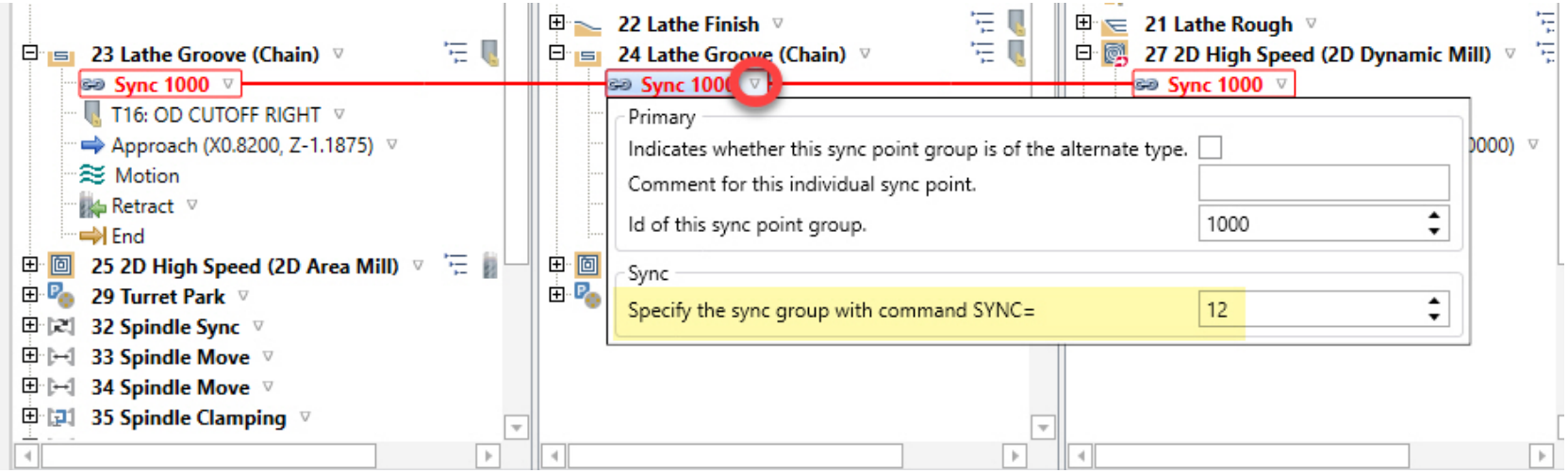
- SYNC=21
- SYNC=12
- SYNC=1

All three commands indicate that turret B is grouped with turret A.

Your post automatically outputs a SYNC=123 command at the beginning of every stream. This means that by default all three turrets are grouped together. Your **.machine** file includes a Sync Manager option that lets you enter your own SYNC commands in any stream as required. To do this, click on the small triangle next to the sync node and enter the SYNC group code as shown.

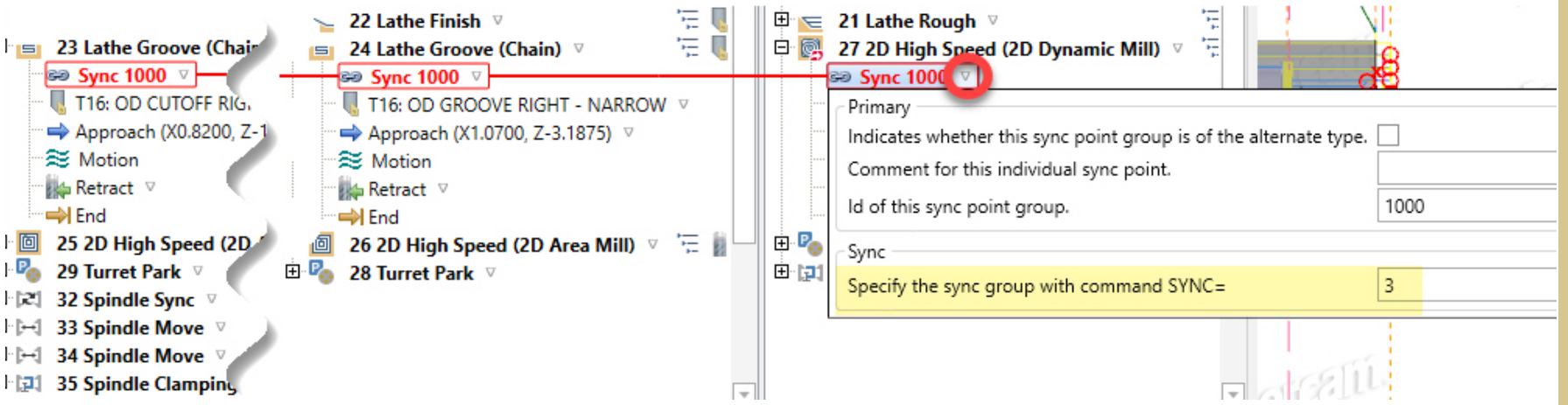


For example, in the following sample part, for sync P1000, turrets A and B are grouped while turret C is working independently. The same SYNC group code is used for turrets A and B:



Home

The code for stream 3 specifies that turret C is working independently:



This results in the following code across the three streams:

```
                                                
```

Using the Load Monitor function

Your Okuma **.machine** file supports Okuma's Load Monitor function. This function is controlled by the `VLMON` command as follows:

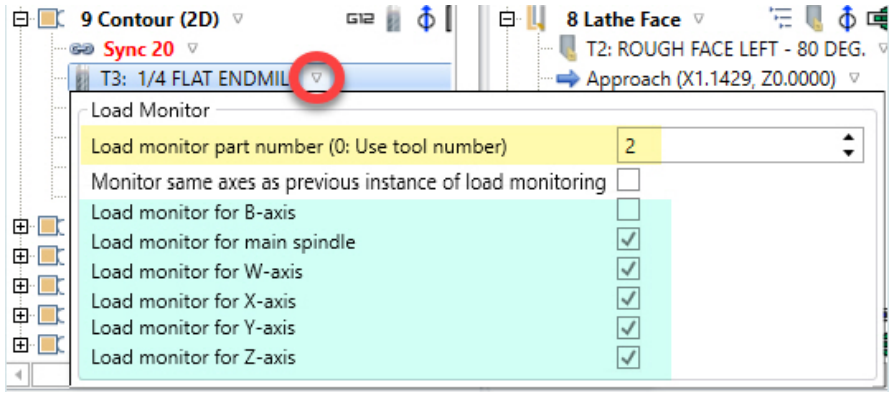
```
VLMON[part number] = monitored axes code
```

where `part number` ranges from 1–64, and the `monitored axes code` is the sum of the individual values for each monitored axis or component, as listed below.

- 0—load monitoring OFF
- 1—X-axis monitoring ON
- 2—Z-axis monitoring ON
- 4—C-axis monitoring ON
- 8—Left spindle monitoring ON
- 16—Tool spindle monitoring ON
- 32—W-axis monitoring ON
- 64— Right spindle monitoring ON
- 128—Y-axis monitoring ON
- 256—B-axis monitoring ON

Use Sync Manager options to turn on and configure Load Monitor. These are attached to the tool node of each operation. For each operation, follow these steps:

1. Click on the small triangle next to the tool name.
2. Enter the desired virtual part number in the **Load monitor part number…** field. Enter 0 to have Mastercam automatically use the tool number, or enter the desired value directly.
3. Select each component to be monitored. Mastercam computes the proper monitored axes code as per the values shown above.



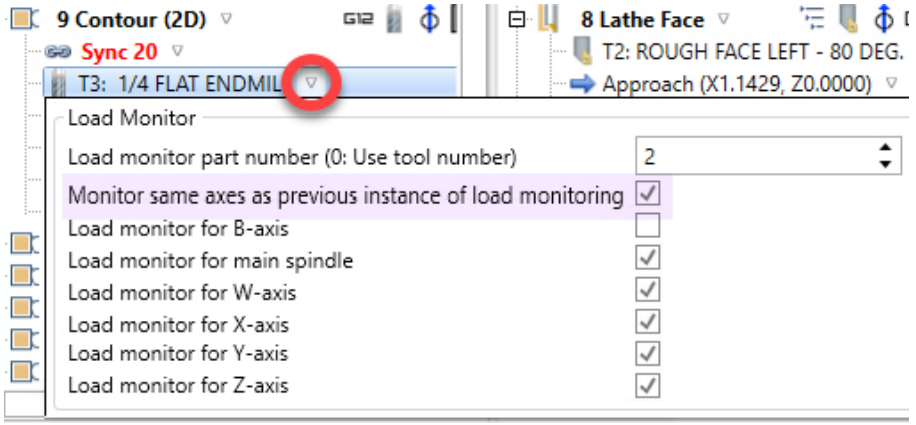
Home

Note that the list of components might be different depending on whether a milling or turning operation is selected, as well as the spindle. For example, if you are machining on the left spindle, the option to monitor the right spindle will not be available.

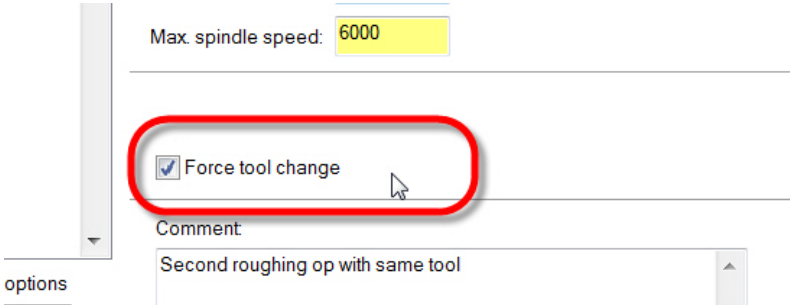
4. Press [**Ctrl+S**] to save the settings to your part file.

Your **.machine** file also lets you reuse the previous load monitor configuration, without needing to reselect each component. Select the **Monitor same axes as previous instance…** to repeat the last used configuration.

- If you select this option, the other options (including the part number) are ignored.
- Note that the previous load monitor instance must be in the same stream.



Since the Load Monitor function is turned on/off at tool changes, the `VLMON` configuration will persist between operations if there is a null tool change. To prevent this, select **Force tool change** in your toolpath settings in Mastercam.



Switching to IJK output for arcs

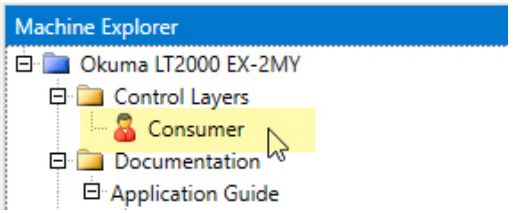
By default your post is configured to output arcs with a radius value `L___` word. If you wish, you can change this to `IJK` output. The following code sample shows the difference:

```
(Operation # 2)
X0.
Z.95
G01 Z.75
G19 G02 Y.75 Z0. L.75
Y0. Z-.75 L.75
Y-.75 Z0. L.75
Y0. Z.75 L.75
G01 Z.95
G00 Z2.
```

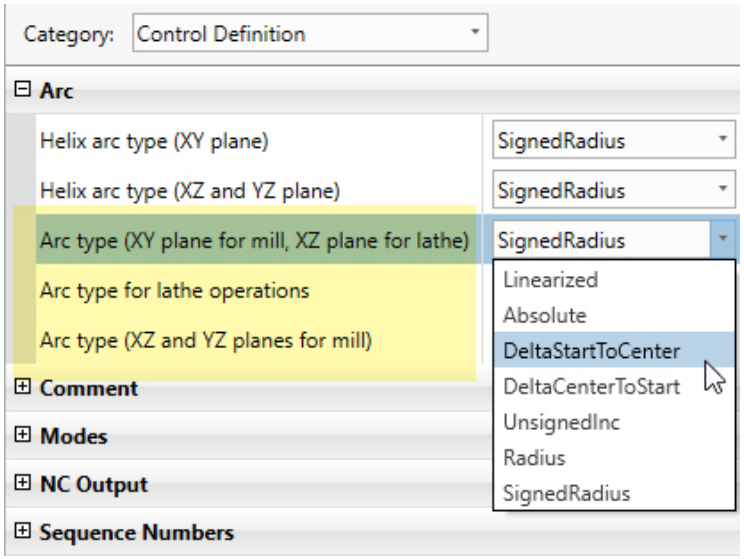
```
(Operation # 2)
X0.
Z.95
G01 Z.75
G19 G02 Y.75 Z0. J0. K-.75
Y0. Z-.75 J-.75 K0.
Y-.75 Z0. J0. K.75
Y0. Z.75 J.75 K0.
G01 Z.95
G00 Z2.
```

Follow these steps to switch to `IJK` output mode:

1. Open the **.machine** file in Code Expert.
2. Double-click the **Consumer** layer.



3. Select **Category: Control Definition**.
4. Go to the **Arc** section.
5. Set the **Arc type…** setting for each type of arc, as desired, to **DeltaStartToCenter** or **DeltaCenterToStart**.
   - Choose **SignedRadius** to restore the default `L___` radius output.
6. Press [**Ctrl+S**] to save the **.machine** file.



Home

D: Configuring part handling operations

Your **.machine** file supports a number of functions that affect how your part handling operations are output.

- Using the G22 torque skip function
- Pickoff at W0 grip position
- Adding a wait code to spindle commands
- Stopping the spindle between operations
- Ejecting the part

Using the G22 torque skip function

Your Okuma **.machine** file supports the `G22` torque skip mode for advancing the subspindle to the grip position.

The torque skip function outputs a `G22` in the following form:

```
G22 Wxx Dxx Lxx Fxx PW=xx
```

where

- `W` is a contact or pickoff location on the part. This is calculated automatically by Mastercam based on your other programming information.
- `D` is the distance before the contact point when the subspindle slows to the feed rate.
- `L` is a limit distance past the contact point to which the subspindle will feed if it has not yet reached the specified load.
- `F` is the feed rate of the subspindle. This is preset in the **.transfer** file by your machine builder and is stored in your **.machine** file.
- `PW` is the desired spindle load as a percentage of maximum load. This is preset by your machine builder and is stored in your **.machine** file.

Home

IMPORTANT: If the D value is greater than the distance between the start and target points, this is an error condition. Mastercam will display an error message if this happens.

The following sections show you how to configure the torque skip function and set the D and L.

G29 is an overall torque limit that is output together with the G22. You can set this yourself and store it in the **.machine** file. G29 is output in the following form:

```
G29 PW=xx
```

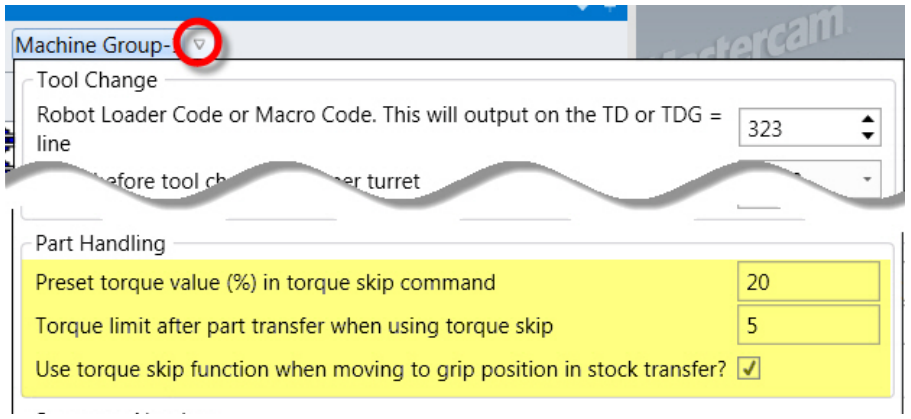
Here is sample output for a typical cycle:

```
(OPERATION # 6)
(PICKOFF SPINDLE - MOVE TO GRIP POSITION)
G29 PW=80
G94 G22 W-31.5 D5. L2. F500. PW=20 (MOVE SECOND SPINDLE TO GRIP POSITION)
G29 PW=5
```

Turning on torque skip mode

Mastercam lets you override the default torque skip setting for each part. Do this in the Sync Manager before you post the part.

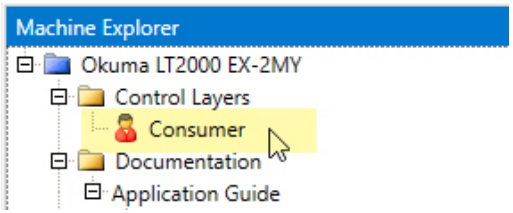
1. In the Sync Manager, click the small triangle next to the machine group name.
2. Select the **Use torque skip function...** option.
3. In the **Preset torque value...** field, enter the desired torque as a percentage of the spindle load. This output as the PW value on the G22 line.
4. Enter the **Torque limit after part transfer...** value. This output as the PW value on the G29 line immediately following the G22 line.
5. Press [**Ctrl+S**] to save the setting back to your part file.



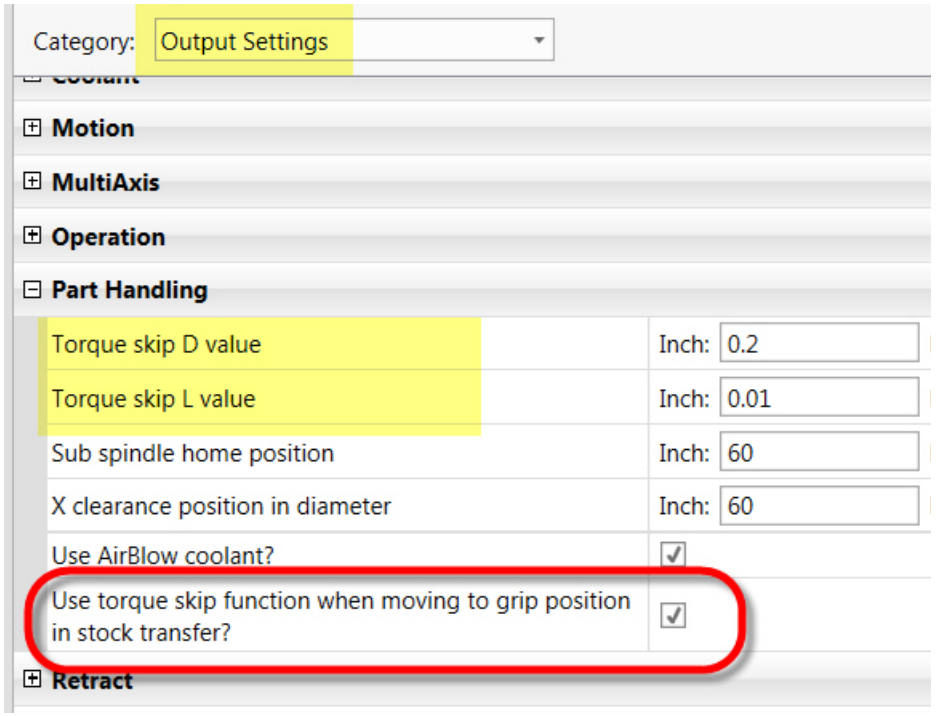
Configuring torque skip mode

Follow these steps to configure torque skip mode.

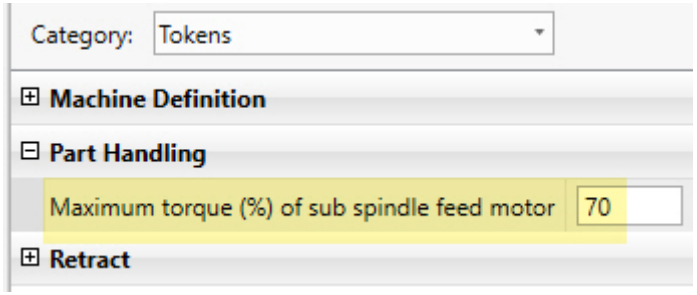
1. Open the **.machine** file in Code Expert.
2. Double-click the **Consumer** layer.



3. Select the **Output Settings** category.
4. Go to the **Part Handling** section.
5. Enter the desired `D` and `L` values in the appropriate fields.
6. Select the **Use torque skip function…** option to have torque skip mode enabled by default for part transfer operations.



7. Select the **Tokens** category.
8. Go to the **Part Handling** section.
9. Enter the **Maximum torque…** value. This will be output with the `G29 PW=` line before the `G22`.
10. Press [**Ctrl+S**] to save the **.machine** file.



Home

Pickoff at W0 grip position

When you are picking off your part during a part transfer operation, by default Mastercam calculates the coordinate of the grip position—the `W` value—from the settings on your **Job Setup** page.

```
(OPERATION # 5)
(PICKOFF SPINDLE - MOVE TO CLEARANCE DISTANCE)
M89 M289
G00 W25. (MOVE SECOND SPINDLE TO CLEARANCE POSITION)

(OPERATION # 6)
(PICKOFF SPINDLE - MOVE TO GRIP POSITION)
G29 PW=30
G22 W-16.392 D5. L2. F500. PW=20 (MOVE SECOND SPINDLE TO
G29 PW=5
```

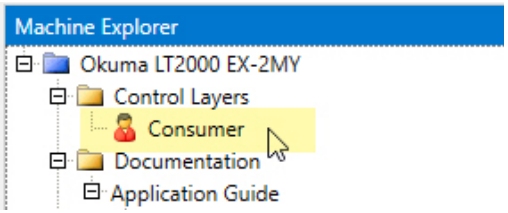
However, your Okuma **.machine** file includes an option that lets you output the grip position as `W0`—regardless of the **Job Setup** settings—if this is how your machine is configured.

```
(OPERATION # 5)
(PICKOFF SPINDLE - MOVE TO CLEARANCE DISTANCE)
M89 M289
G00 W41.392 (MOVE SECOND SPINDLE TO CLEARANCE POSIT

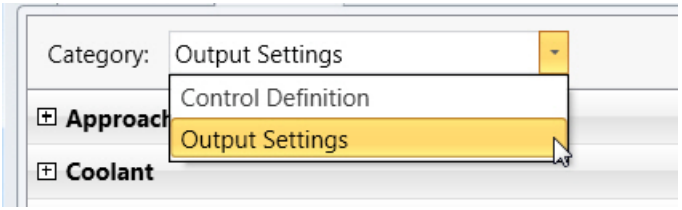
(OPERATION # 6)
(PICKOFF SPINDLE - MOVE TO GRIP POSITION)
G29 PW=30
G22 W0. D5. L2. F500. PW=20 (W0 IS GRIP POSITION)
G29 PW=5
```

Follow these steps to select the `W0` option:

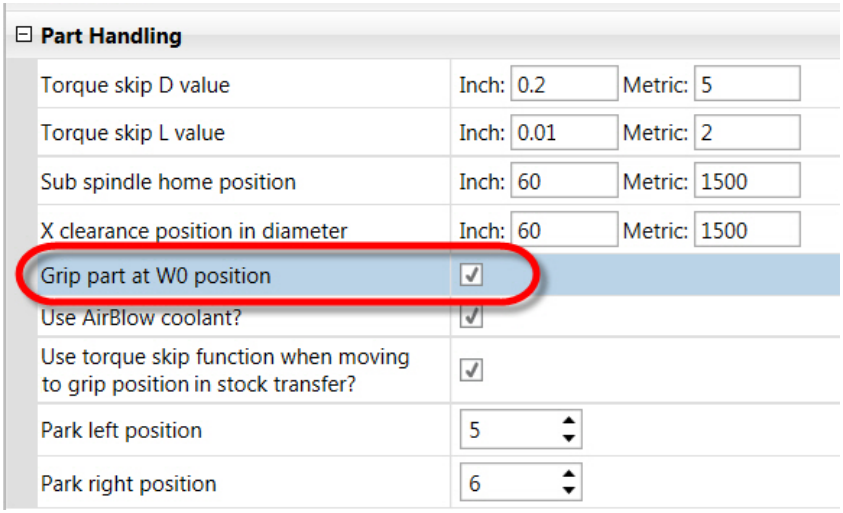
1. Open the **.machine** file in Code Expert.
2. Double-click the **Consumer** layer.



3. Select the **Output Settings** category.



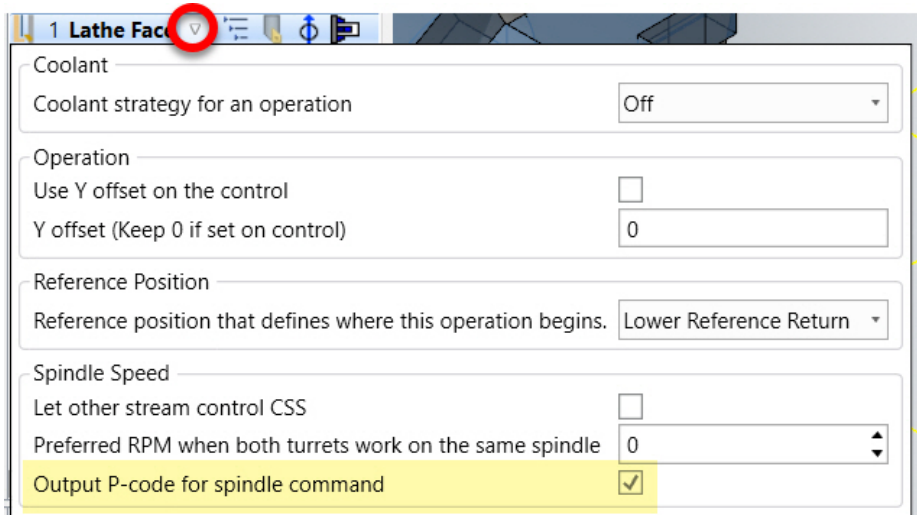
4. Go to the **Part Handling** section.
5. Select the **Grip part at W0 position...** option.
6. Press [**Ctrl+S**] to save the **.machine** file.



Adding a wait code to spindle commands

If your Okuma machine has two turrets, you might choose to output a wait code (for example, `P1` or `P2`) with spindle speed or spindle off commands. These codes are not synced with matching codes in the other stream. Your **.machine** file includes an option for adding them to each operation as desired. Follow these steps:

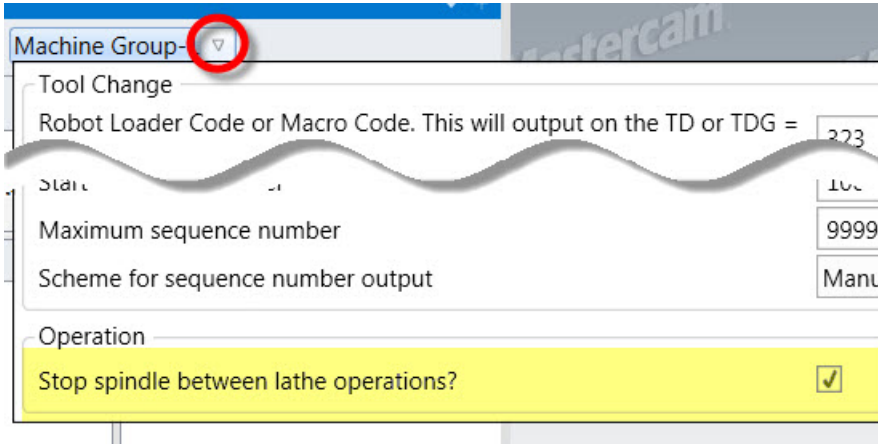
1. Click the small triangle next to the operation name.
2. Select the **Output P-code for spindle command** option. It is available for all turning operations.
3. Press [**Ctrl+S**] to save the setting back to your part file.



Stopping the spindle between operations

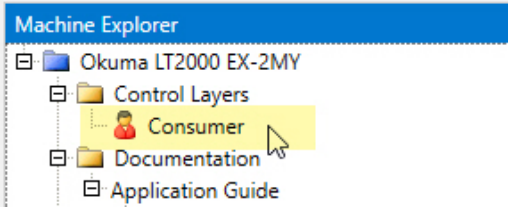
You can configure your **.machine** file to force a spindle stop code between each operation.

Click the small triangle next to the machine group name and select the **Stop spindle between lathe operations** option.

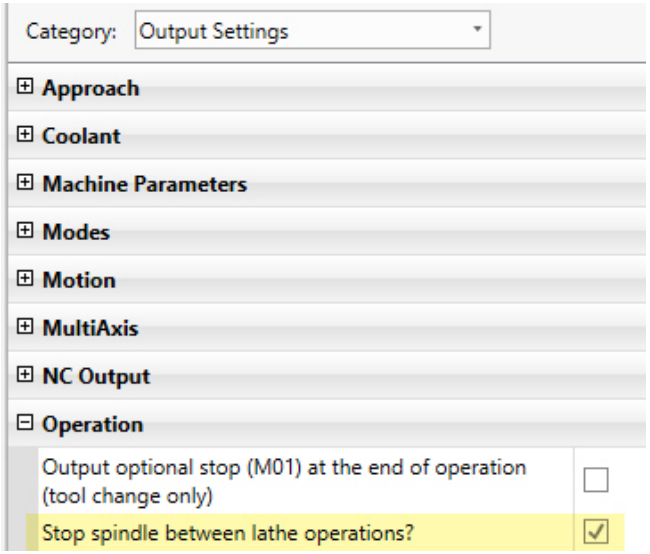


You can also choose the default mode. Follow these steps:

1. Open the **.machine** file in Code Expert.
2. Double-click the **Consumer** layer.



3. Select **Category: Output Settings**.
4. Go to the **Operation** section.
5. Select the desired mode.
6. Press [**Ctrl+S**] to save the **.machine** file.



Home

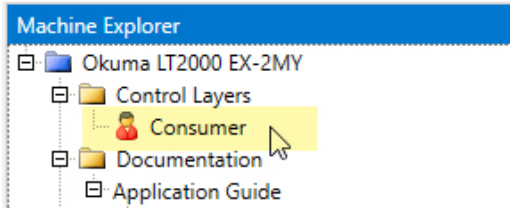
Ejecting the part

Your **.machine** file gives you several options for unloading or ejecting your part at the end of your NC program.

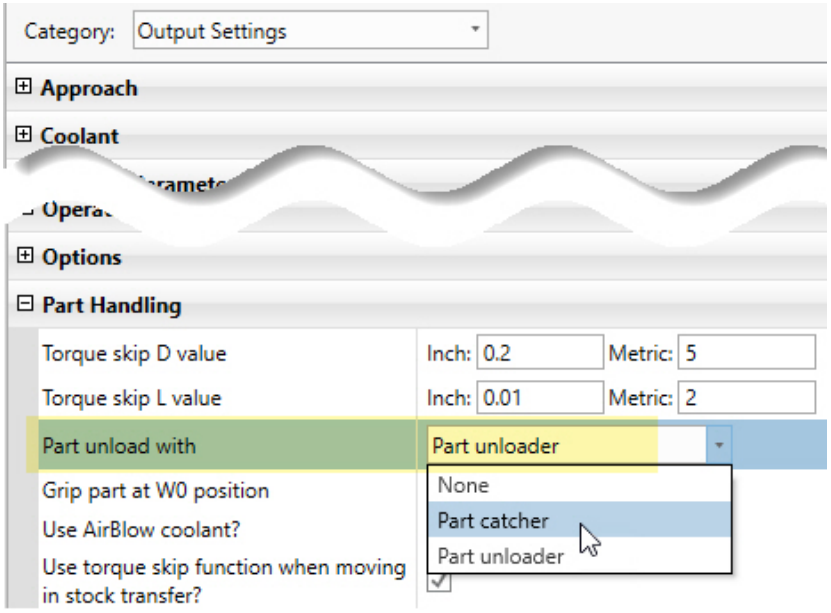
- Output a complete part unload sequence.
- Use a part catcher (`M77/M76`).
- None.

Follow these steps:

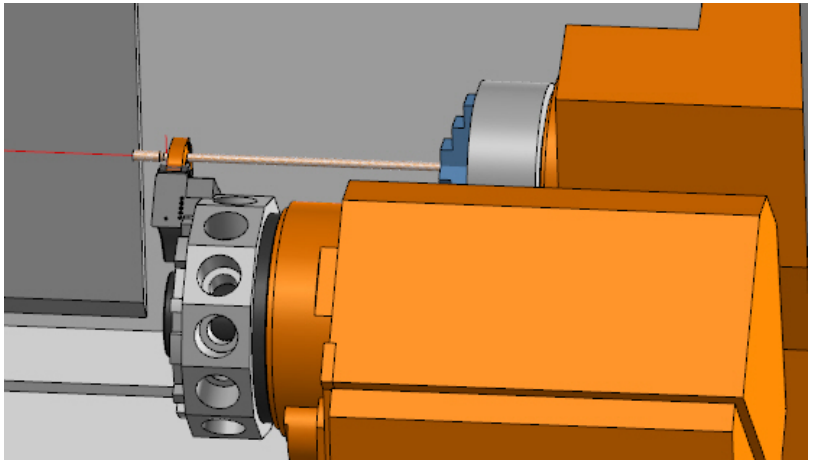
1. Open the **.machine** file in Code Expert.
2. Double-click the **Consumer** layer.



3. Select **Category: Output Settings**.
4. Go to the **Part Handling** section.
5. Select the desired **Part unload with** option.
6. Press [**Ctrl+S**] to save the **.machine** file.



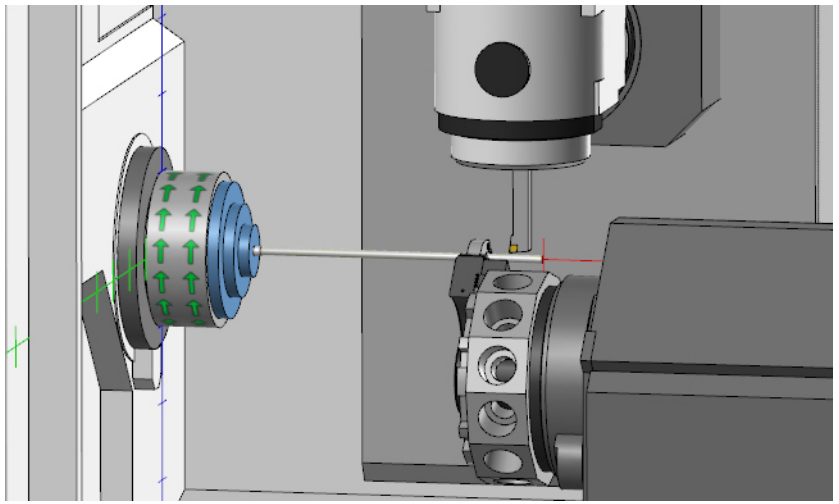
E: Steady rest and tailstock functions



Many Okuma **.machine** files include support for a turret-mounted steady rest. The following topics explain how to support several common steady rest applications:

- **Following the subspindle home with the steady rest**
- **Following the tool with the steady rest**
- **Steady rest options (M472, M102)**
- **Supporting tailstock operations**

Following the tool with the steady rest



Mastercam lets you create steady rest operations that follow a tool or the other turret while the part is being machined. However, there are several guidelines that you need to follow. The following outline summarizes the steps and requirements.

1. Create a steady rest operation that clamps the steady rest. Choose a location on your part where the diameter will not change while the steady rest follows the part. The follow operation will not be successful if the diameter changes.

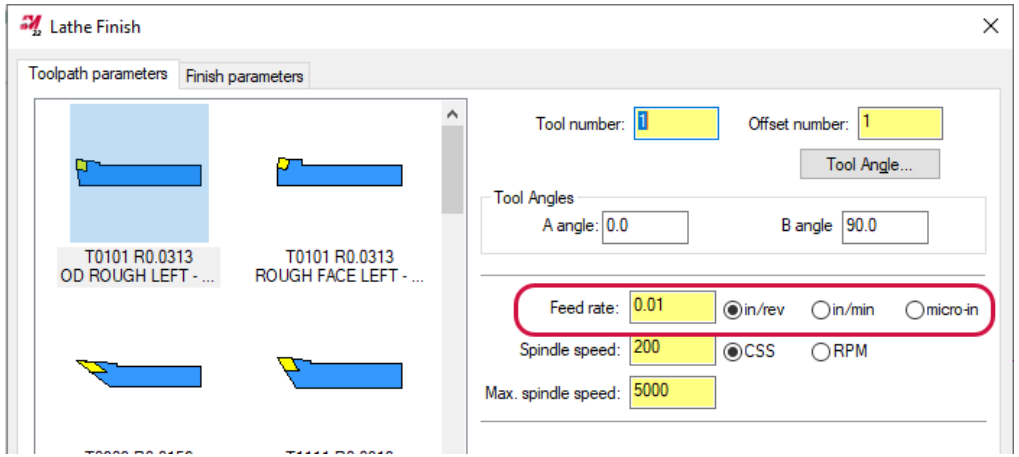
   To be clear, the turning operation can include sections of the part with different diameters, but the portion of the part over which the steady rest travels must be a constant diameter.

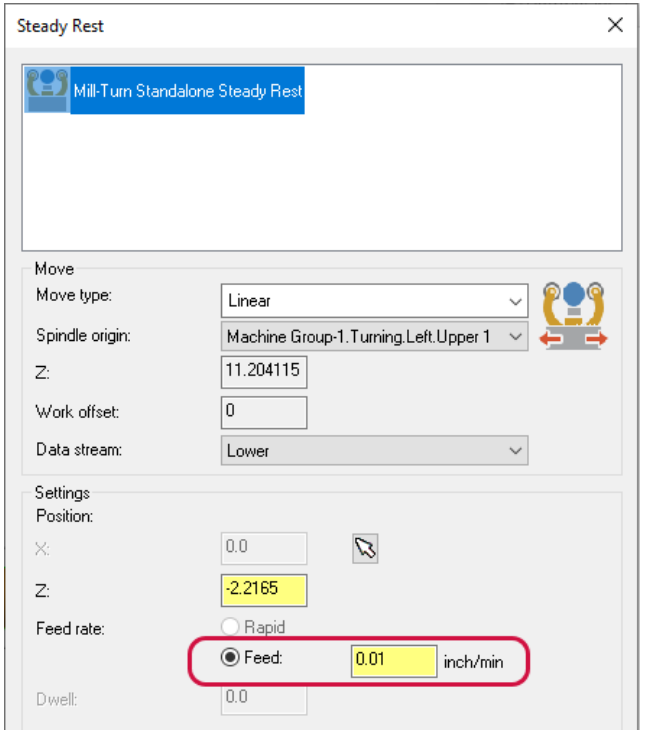
2. Once the steady rest is clamped, create the turning operation. The only supported operations are rough and finish turning operations.

3. Create the steady rest operation that will follow the turning operation. Use only linear or feed rate moves, not rapids.

Home

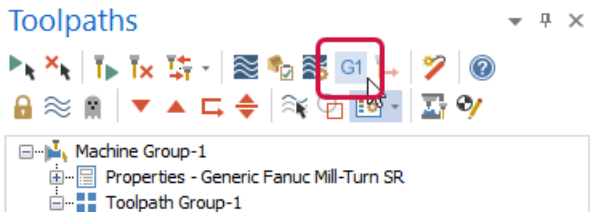
4. Ensure that the feed rates for the turning operation and the steady rest are the same. The picture at the right shows an operation for a standalone steady rest, but this requirement is the same for both types of steady rest operation.





Note— When you create the turning toolpath, Mastercam lets you specify the units for the feed rate, but not for the steady rest operation. You only need to ensure that the number matches; it does not matter if the units are different. When the steady rest and turning operations are properly synced (see step 6 below), Mastercam will adjust for the different units in your post and your code will be output correctly.

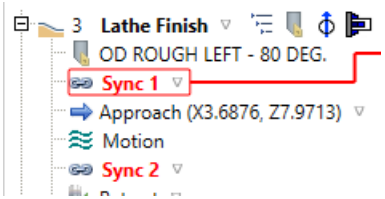
5. When you have finished creating operations, click the **G1** button to post your operations to the Sync Manager.

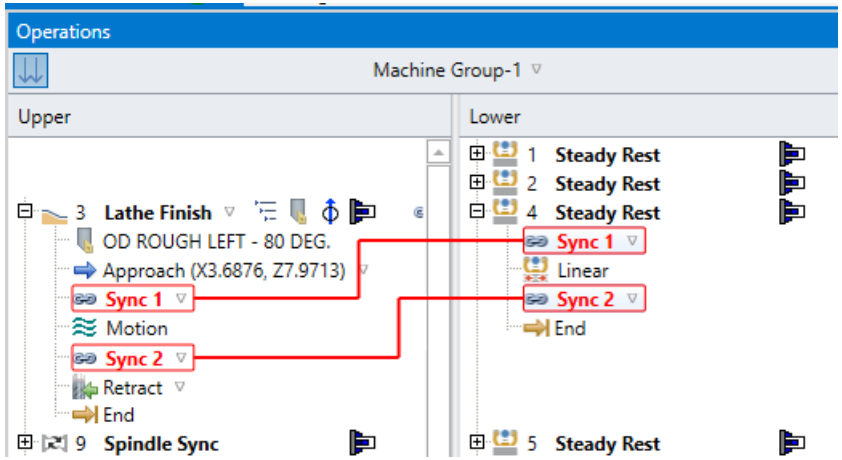


Home

6. In the Sync Manager, sync the turning operation and the steady rest follow operation as shown here. Create one sync between the beginning of the steady rest operation and the beginning of the motion block in the turning operation, and another sync between their ends.

   Notice that the first sync must be placed before the **Motion** block, not the **Approach** move:





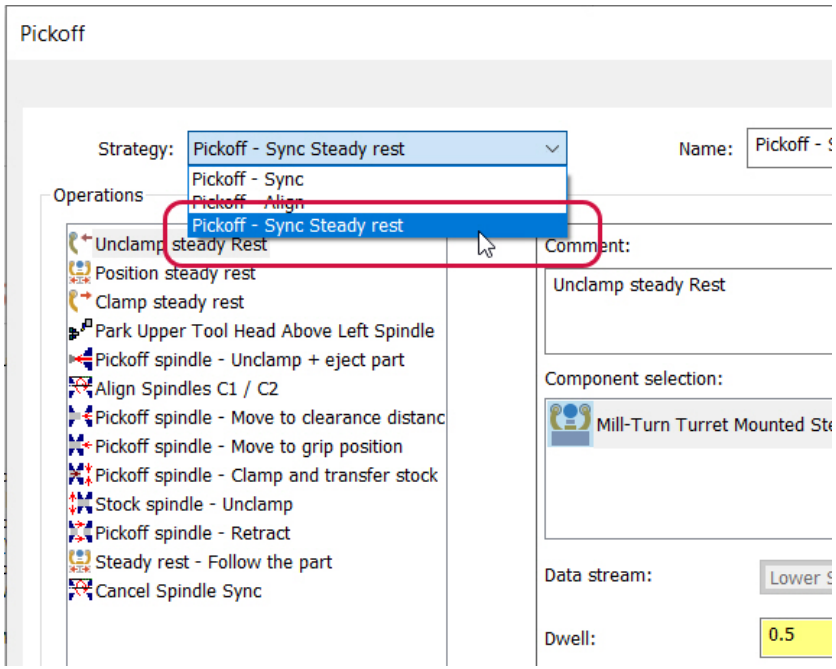
You can only follow one operation at a time. For each turning operation that you want to follow, you need to create a separate steady rest operation.

Home

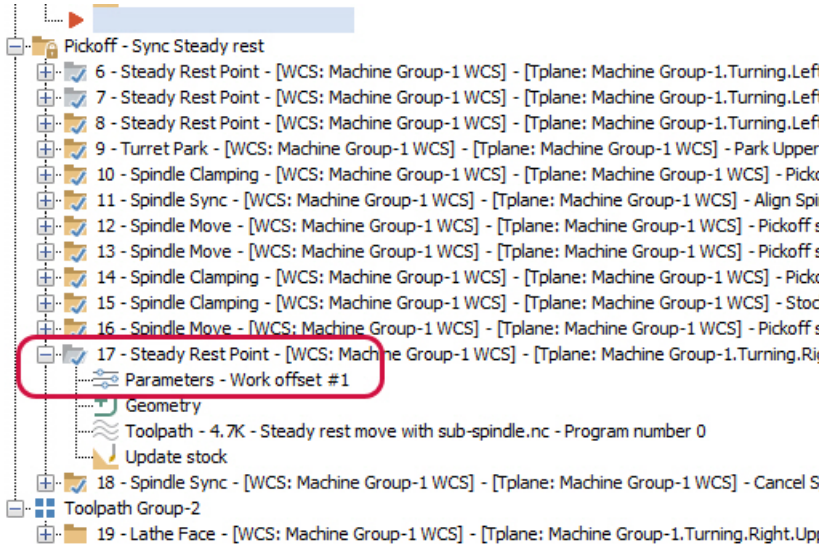
Following the subspindle home with the steady rest

Mastercam lets you create steady rest operations that follow the subspindle home while supporting the part after a pickoff by the subspindle. To do this, select a part handling strategy that includes steady rest operations. The following outline summarizes the steps and requirements.

1. Select a part handling strategy the includes steady rest operations. If your **.machine** file includes such strategies, they will be clearly labeled in the **Strategy** list:

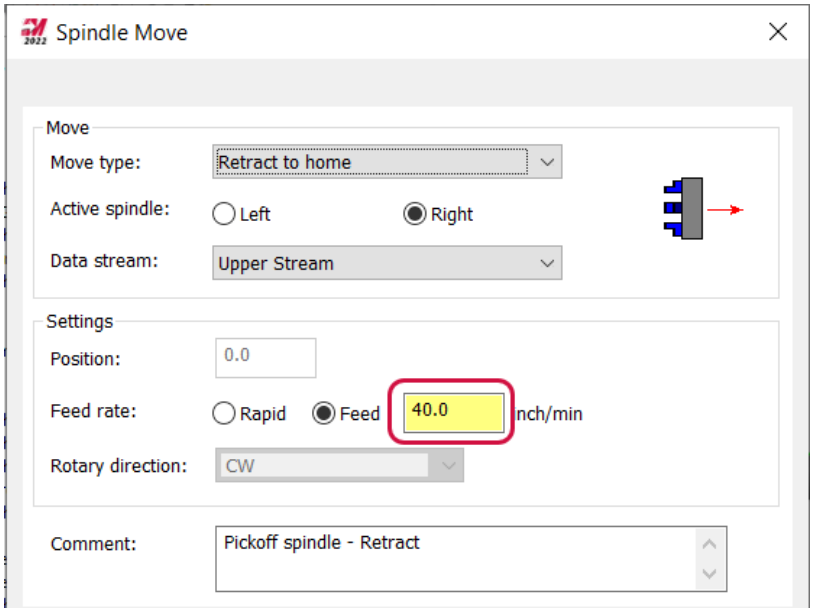


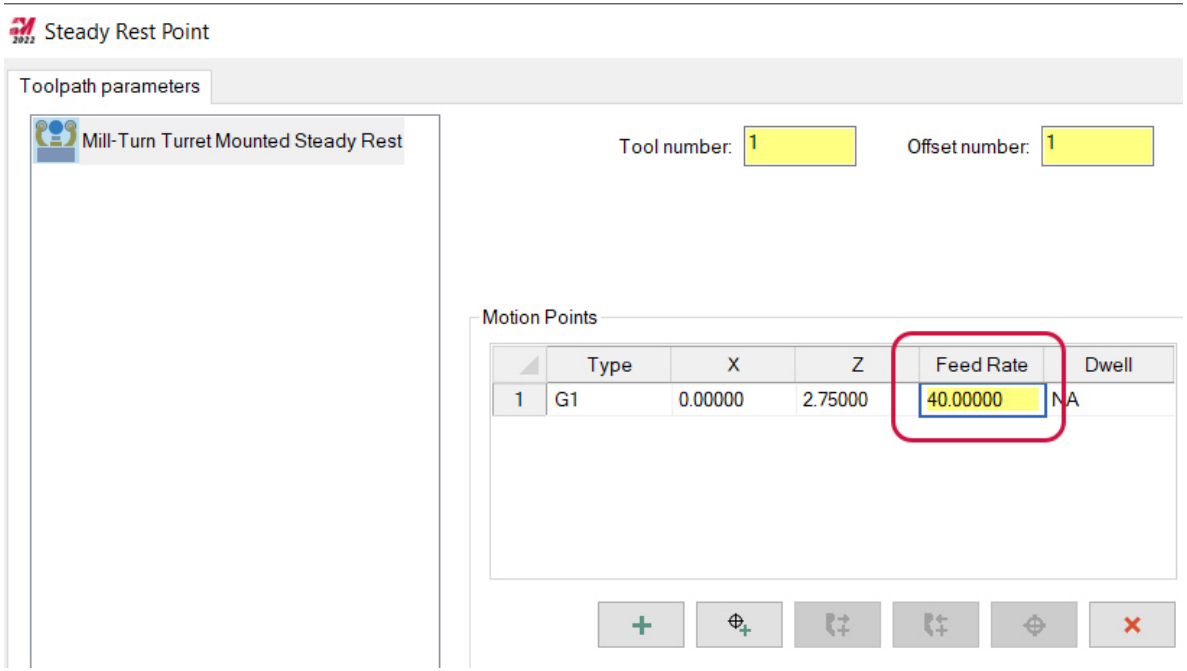
2. After Mastercam has generated the part handling operations, click the **Parameters** icon for the final steady rest operation. In this example, it is operation 17.



Home

3. Ensure that the feed rate for the steady rest operation is the same as the feed rate for the spindle move operation that immediately precedes it.



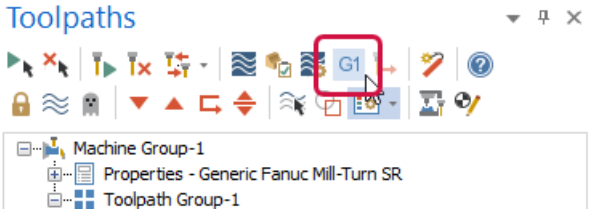


Note— Do not worry if the feed rate units are not the same between the two operations.You only need to ensure that the number matches; it does not matter if the units are different. When the steady rest and turning operations are properly synced (see step 6 below), Mastercam will adjust for the different units in your post and your code will output correctly.

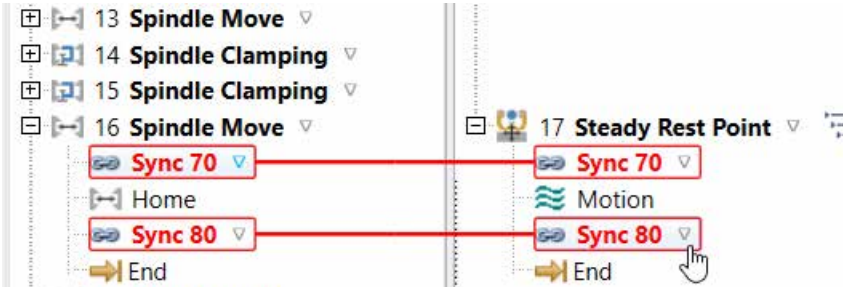
4. Ensure that the **Lower Right** axis combination is selected for the steady rest operation.



5. When you have finished creating operations, click the **G1** button to post your operations to the Sync Manager.



6. In the Sync Manager, sync the steady rest follow operation to the spindle operation in which the subspindle returns home. Create one sync between the beginning of the steady rest operation and the beginning of the spindle operation, and another sync between their ends.

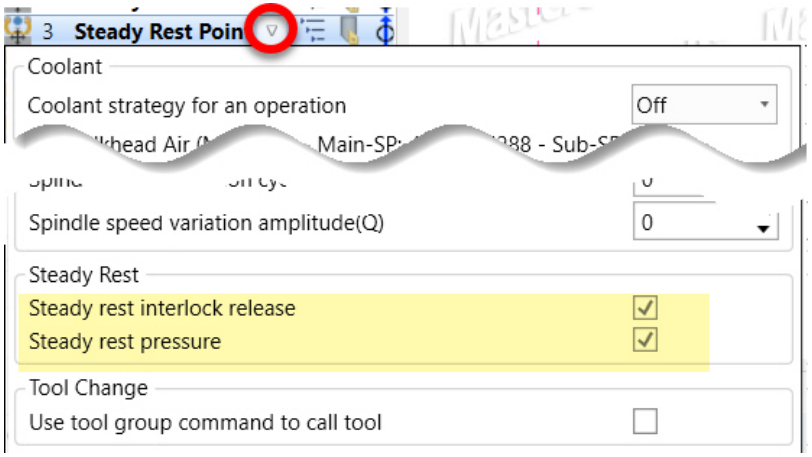


Steady rest options (M472, M102)

Your **.machine** file includes two additional steady rest options:

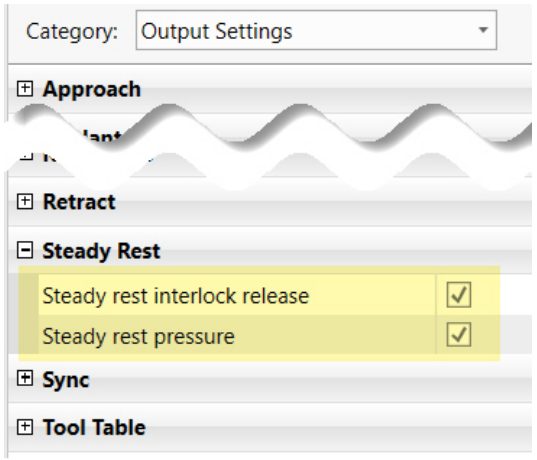
- Turn on steady rest interlock mode (`M472`)
- Turn on steady rest clamping pressure control (`M102`)

Click the small triangle next to any steady operation to access either option.



You can also turn on either mode by default. Follow these steps:

1. Open the **.machine** file in Code Expert.
2. Double-click the **Consumer** layer.
3. Select the **Output Settings** category.
4. Go to the **Steady Rest** section.
5. Select either option as desired to enable it by default.
6. Press [**Ctrl+S**] to save the **.machine** file.

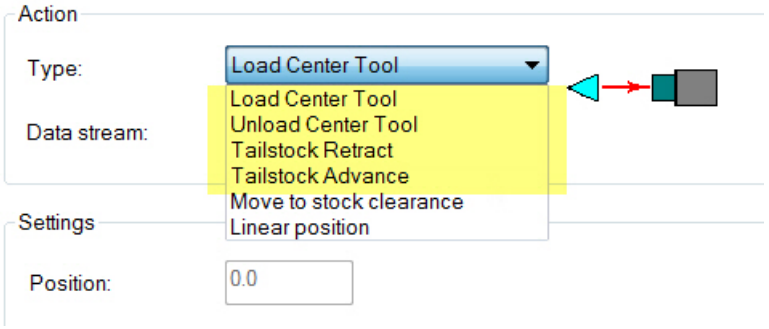


Home

Supporting tailstock operations

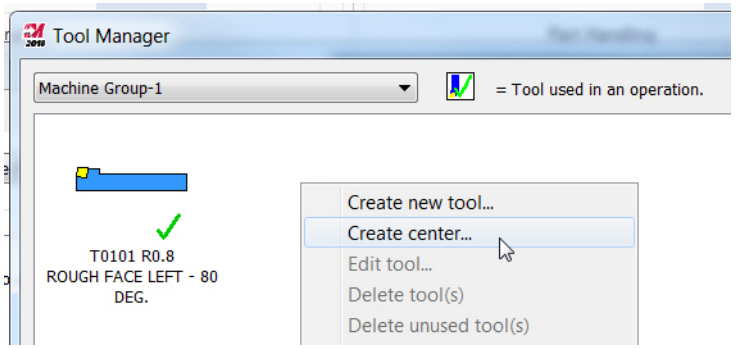
Several Okuma **.machine** files include tailstocks. Even though tailstock support is built into the Mastercam Mill-Turn product itself, you cannot create tailstock operations unless your **.machine** file includes the proper tailstock components and other supporting resources.

Supported tailstock operations—The **Move to stock clearance** and **Linear position** operation types are not supported by your **.machine** file. If you select either of these operation types, you will get an error when you post. Only the tailstock operations highlighted at right are supported.

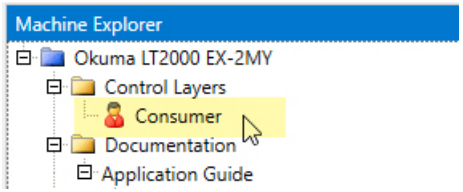


Creating a center tool—In Mastercam Mill-Turn, tailstock centers are considered a type of tool and are stored in your tool library.

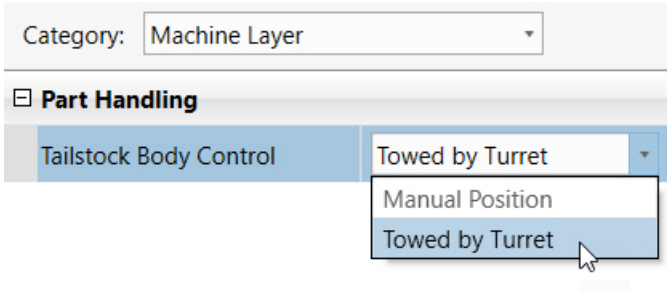
Right-click in the **Tool Manager** and select **Create center** if you need to create one.



Configuring the tailstock motion mode—Your **.machine** file includes an option that you can use to tell the post whether your tailstock is positioned manually or is towed by the turret. Follow these steps:



1. Open the **.machine** file in Code Expert.
2. Double-click the **Consumer** layer.

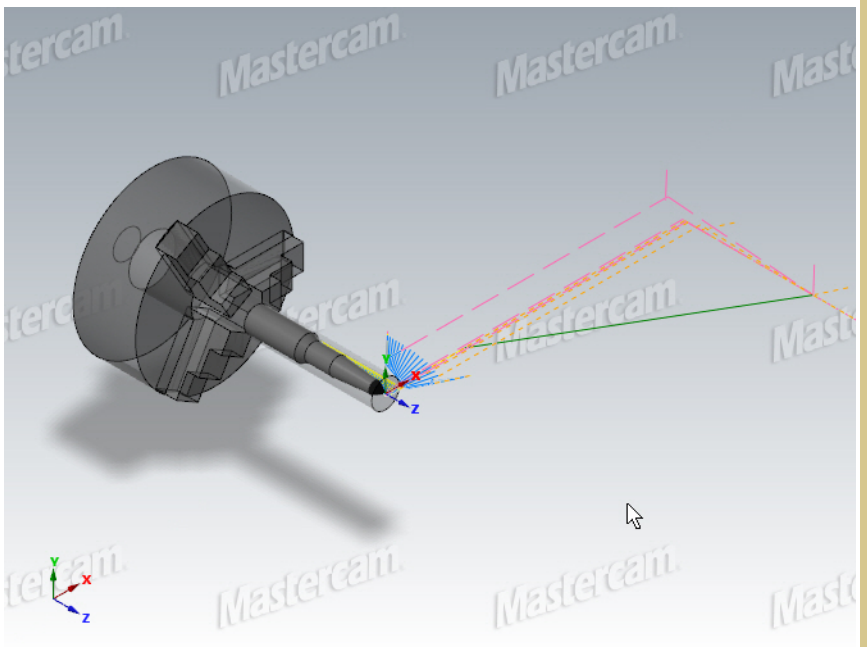


3. Select **Category: Machine Layer**.
4. Go to the **Part Handling** section.
5. Select the desired mode.
6. Press [**Ctrl+S**] to save the **.machine** file.

F: Approach and retract moves

Your **.machine** file gives you several options for configuring approach and retract moves:

- You can select the start point for the approach move and the end point of the retract. These are specified with reference positions. See **Selecting reference positions**.
- You can choose the type of motion for each: dogleg (Z-first or X-first) or a direct move. See **Setting the type of approach/retract motion**.
- You can use variables to control the home position destinations for turret park moves. See **Using variables to set home positions**.



Home

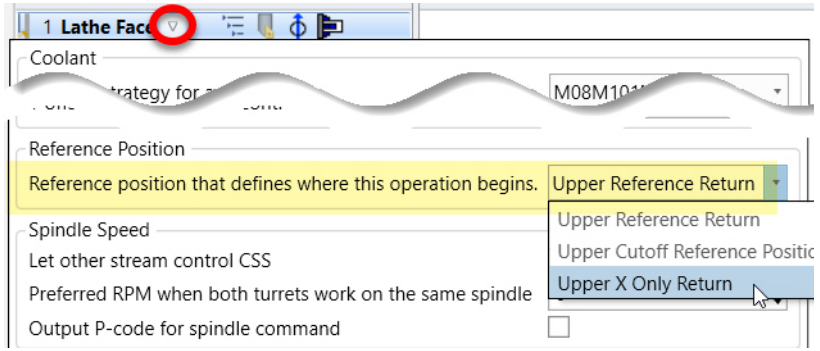
Selecting reference positions

Use reference positions to tell Mastercam where you want to start and end each operation. You can select these positions individually for the start point and end point of each operation. For example, you might—or might not—want to move all the way to the home position between operations.

You can also define additional, new reference positions. Do this in the Job Setup inside Mastercam. Do this if your particular part setup requires different reference locations than are already defined in your **.machine** file—for example, to accommodate special fixturing, an unusual part shape, etc.

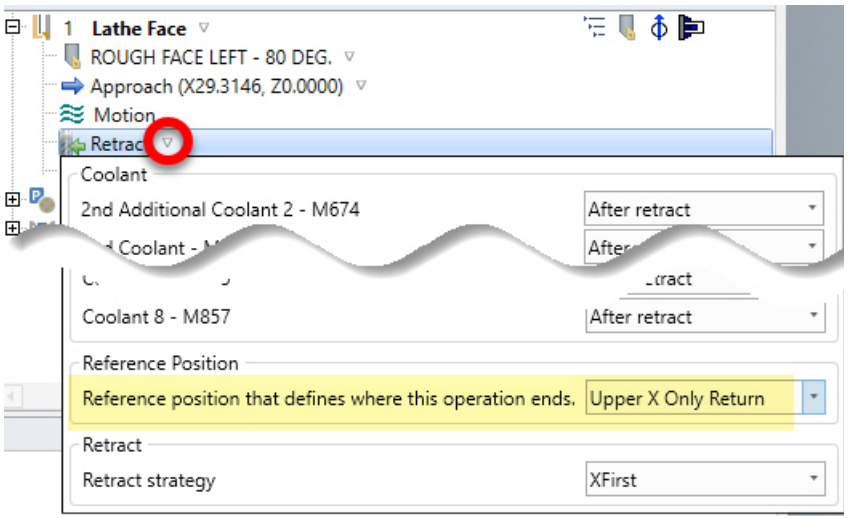
Setting the start point for an operation

To set the start point for an operation, select the desired **Reference position** in the Sync Manager. Click the small triangle next to the operation name in the Sync Manager, and select the location from the list.



Setting the end point

To set the end point for an operation, select the desired **Reference position** from the operation's **Retract** branch. You can choose to make the end position of one operation the same location as the start point of the next operation by choosing the same reference position.



Selecting a reference position of None

Selecting **None** for a reference position means that there will simply be no output where the reference position is typically output. For example, the code at right shows what happens if **None** is selected as the reference position for operation #2.

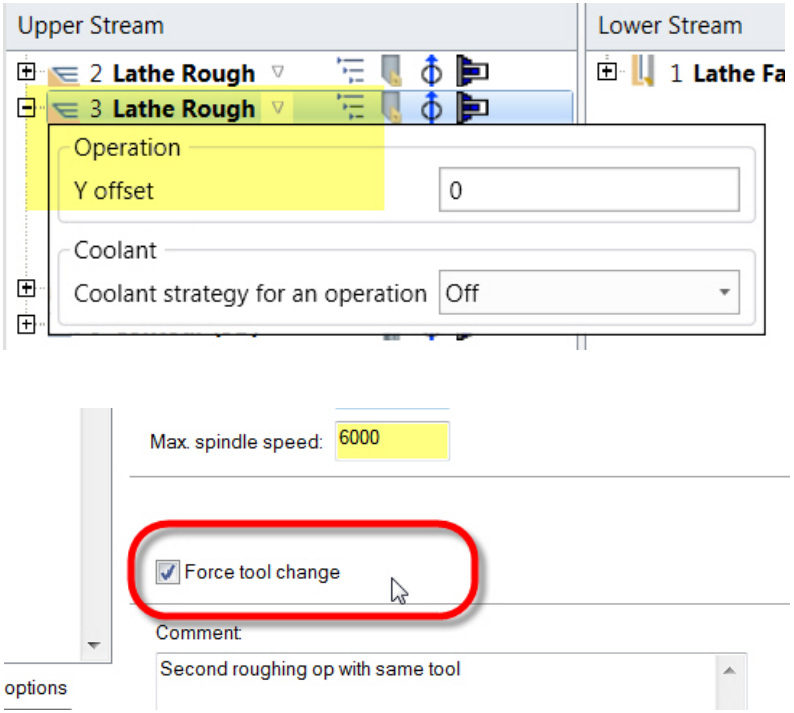
Note that you cannot select **None** for the start point of the first toolpath.

```
54   (Operation # 2)
55   (2  |  OD 55 deg Left  | )
56   NP002
57   M321
58   MT=101
59   TL=020202 BT=0 BA=90. G52 M602
60   NT002
61   G18
62   G97 S491 M04 M41
```

Reference positions and null tool changes

If you have consecutive operations in the same stream that use the same tool and tool orientation, Mastercam will typically not output a tool change between the operations. When this happens, Mastercam will not display the reference position option for the **Retract** of the first operation, or for the start of the second operation. In the picture at right, operations 2 and 3 use the same tool. You can see that the **Reference position** option is not available for operation 3.

In these instances, you can force the **Reference position** option to be available by selecting the **Force tool change** option inside Mastercam. This might be useful if you have defined custom reference positions that you want to use as clearance positions between such operation.

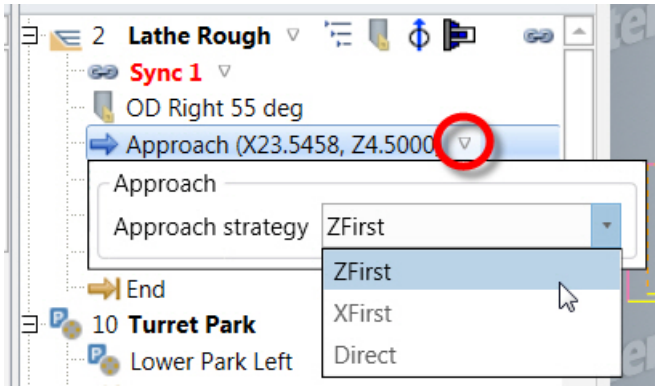


Setting the type of approach/retract motion

For each approach and retract move, you can select the following motion:

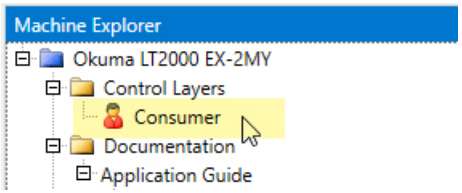
- X-first
- Z-first
- Direct (interpolated) move

Click the small triangle next to the **Approach** or **Retract** node in the tree.

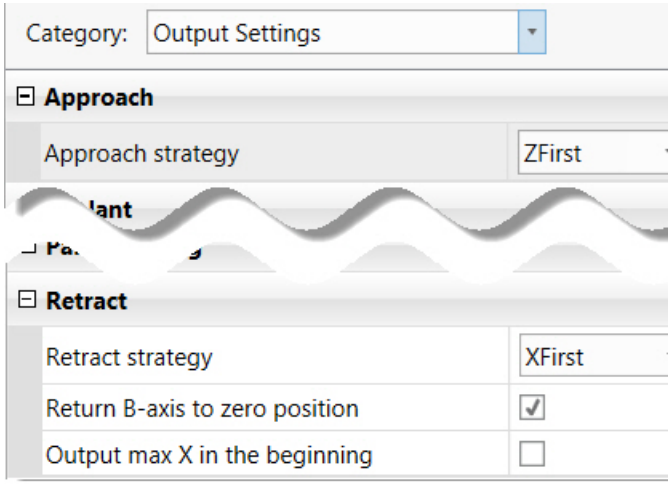


You can also change the default selection in the **.machine** file.

1. Double-click the **Consumer** layer.
2. Go to the **Output Settings** category.



3. Open the **Approach** and **Retract** group.
4. Select the desired strategy for each.
5. Press [**Ctrl+S**] to save your settings.



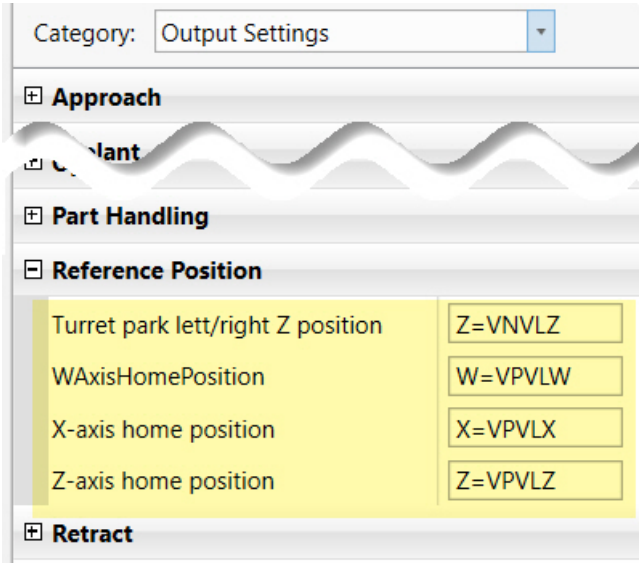
Using variables to set home positions

The post in your **.machine** file is written to use system variables to set the following home or reference return positions:

- Turret park left and right Z-coordinate (VNLVZ)
- W-axis reference return/home position (VPVLW)
- X-axis reference return/home position (VPVLX)
- Z-axis reference return/home position (VPVLZ)

If you wish, you can edit the NC code or variable names associated with these commands, or replace them with actual coordinate positions. Follow these steps:

1. Open the **.machine** file in Code Expert.
2. Double-click the **Consumer** layer.
3. Select **Category: Output Settings**.
4. Go to the **Reference Position** section.
5. Enter the variable assignments or names as shown here. The entire string that you enter in each field will be output exactly as you type it here.
6. Press [**Ctrl+S**] to save the **.machine** file.



The following code sample shows how the strings will be output:

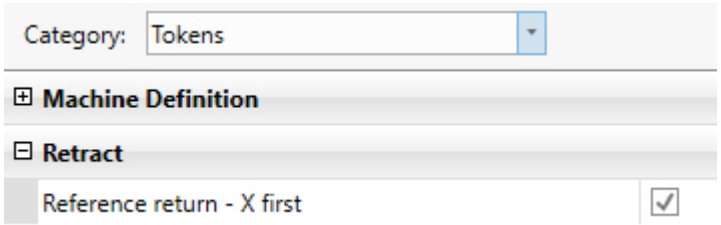
```
G140
G00 W=VPVLW (MOVE SUB SPINDLE HOME)
G50 S4500 (MAX SPINDLE SPEED FOR MAIN SPINDLE)
G141
G50 S4500 (MAX SPINDLE SPEED FOR SUB SPINDLE)
P-10

G13
G140
(Operation # 1)
N4 (4   |  ROUGH FACE LEFT - 80 DEG.  | )
NP001
M109
G00 X=VPVLX (X AXIS RETURN HOME)
G00 Z=VPVLZ (Z AXIS RETURN HOME)
NA04 T40404
G18
G97 S259 M04 M41 P1
G95 G00 Z0.
X2.9465
G96 G110 S200 P2
G95 G01 X-.2268 F.01
G00 Z.1
G97 S259 P3
M05 P4
G00 X=VPVLX (X AXIS RETURN HOME)
G00 Z=VPVLZ (Z AXIS RETURN HOME)
```

Configuring reference return moves

Your **.machine** file lets you choose whether reference return moves are output X-first or Z-first. Follow these steps:

1. Open the **.machine** file in Code Expert.
2. Double-click the **Consumer** layer.
3. Select **Category: Tokens**.
4. Go to the **Retract** section.
5. Select the **Reference return - X first** option if desired. If this is not selected, the Z move will be output first.
6. Press [**Ctrl+S**] to save the **.machine** file.



G: Configuring milling operations

Your **.machine** file supports the following milling cycles:

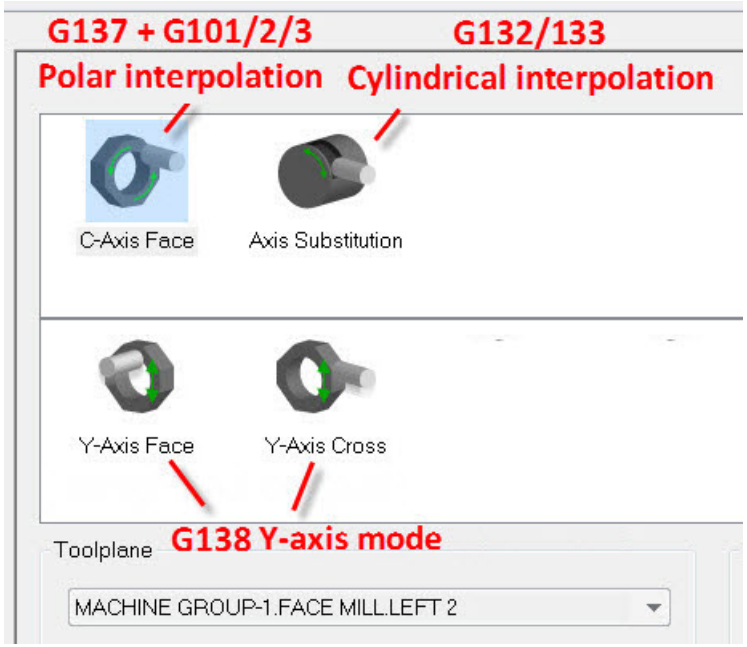
- **Cylindrical interpolation (G132/G133)**
- **Polar interpolation (G137) and contour generation (G101/G102/G103)**

The cycles are automatically keyed to the setup types on the **Setup** page for mill toolpaths. The appropriate cycle will be activated in your post based on the setup type that you select.

Before you program milling operations, complete the steps in **Enabling live tooling** on page 45.

See the following sections to learn more about additional features:

- **Y-axis (G138) mode**
- **C-axis clamping (M147)**
- **Spindle commands for pinch/balanced milling operations**
- **C-axis rotation direction**

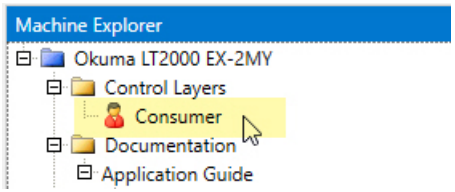


Home

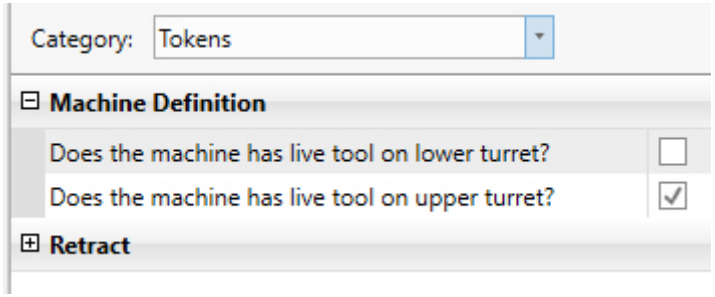
Enabling live tooling

Before you use your **.machine** file to create milling operations, you need to tell Mastercam which turrets will be used for live tooling. Follow these steps:

1. Open the **.machine** file in Code Expert.
2. Double-click the **Consumer** layer.



3. Select **Category: Tokens**.
4. Go to the **Machine Definition** section.
5. Select the turrets for which you intend to use live tooling.
6. Press [**Ctrl+S**] to save the **.machine** file.



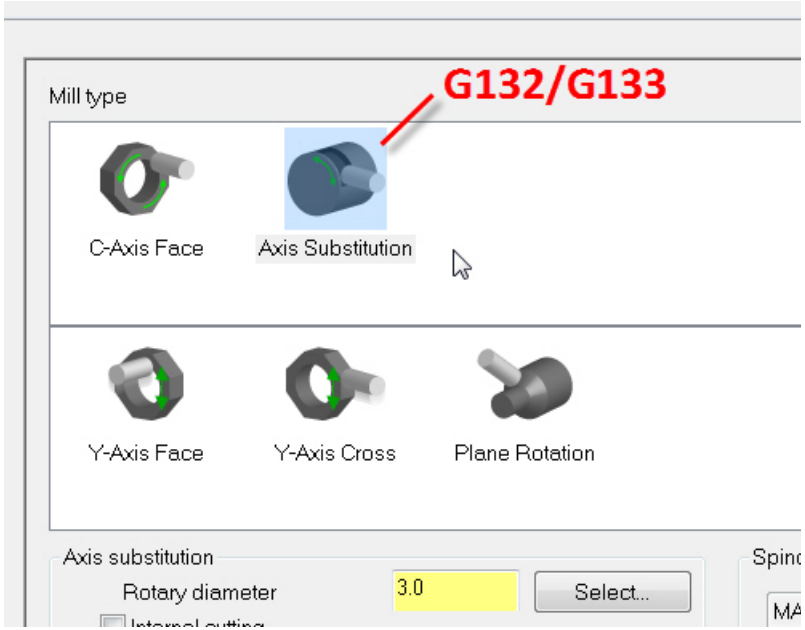
Home

Cylindrical interpolation (G132/G133)

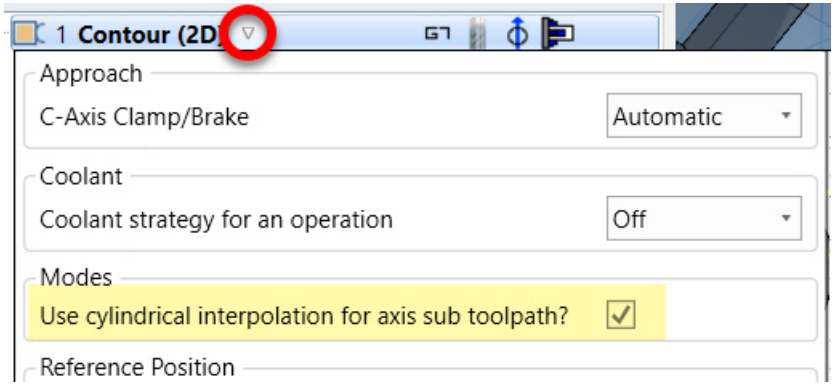
Your **.machine** file supports cylindrical interpolation cycles (G132/G133) for axis substitution toolpaths. The post automatically selects G132 or G133 based on which axis needs to be substituted.

Follow this general workflow for programming axis substitution toolpaths.

1. The G132/G133 cycles are automatically enabled when you select **Axis Substitution** on the **Setup** page for milling operations.
2. The **Toolplane** is automatically selected. Enter the **Rotary diameter** and other options as appropriate for your selected geometry.

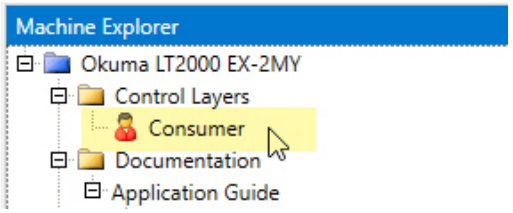


3. For each operation, choose whether or not to use G132/G133 cylindrical interpolation. Click the small triangle next to the operation and select the cycle as shown.
4. Press [**Ctrl+S**] to save the setting.

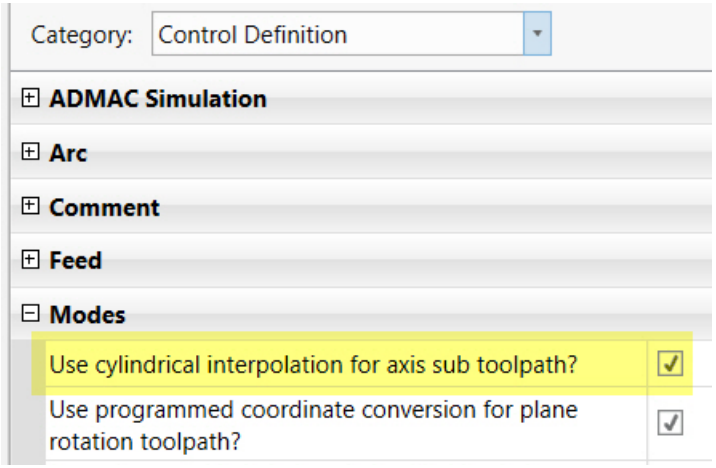


You can also enable or disable the cycle by default. Follow these steps:

1. Open the **.machine** file in Code Expert.
2. Double-click the **Consumer** layer.

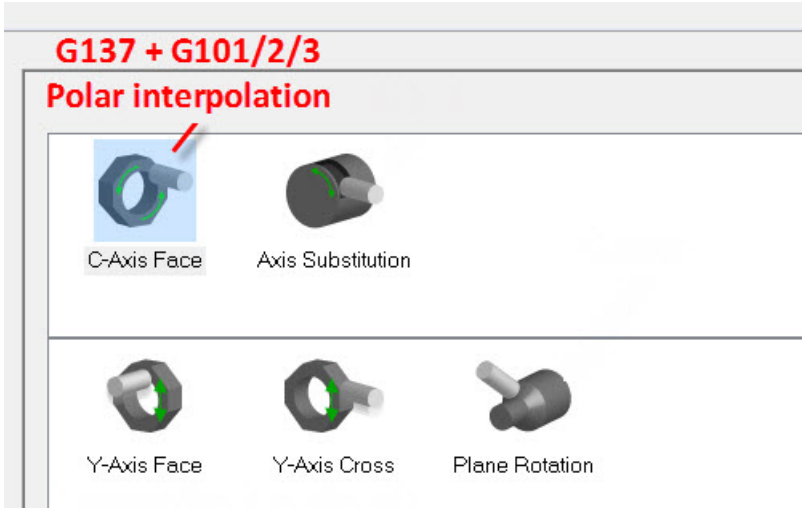


3. Select **Category: Control Definition**.
4. Go to the **Modes** section.
5. Select the **Use cylindrical interpolation…** option to use `G132/G133` by default.
6. Press [**Ctrl+S**] to save the **.machine** file.



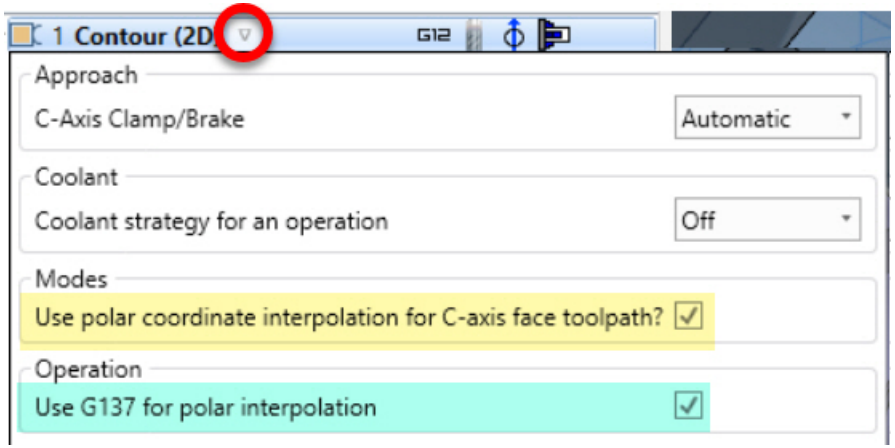
Polar interpolation (G137) and contour generation (G101/G102/G103)

Your P300 **.machine** file supports both polar interpolation (`G137`) and the `G101/G102/G103` contour generation mode for C-axis face toolpaths. Follow this general workflow.



1. Select **C-axis Face** on the **Setup** page when creating the operation in Mastercam. This enables the polar cycles.

   The **Toolplane** is automatically selected when you select **C-axis Face**. These planes were created for you when you completed the **Job Setup** process. Each plane sets the proper tool orientation for your part.

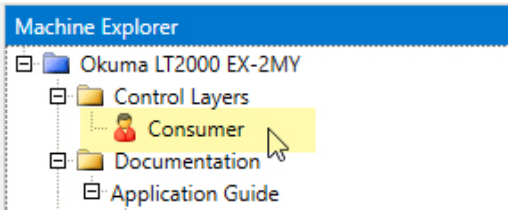


2. In the Sync Manager, click the small triangle next to the operation.
3. Select **Use polar coordinate interpolation…** (the yellow option) to enable the `G101/G102/G103` contour generation mode. You must select this option if you want to use `G137`.
4. Select **Use G137 for polar interpolation** (the green option) to output a `G137` command.
5. Press [**Ctrl+S**] to save the settings back to your part file.

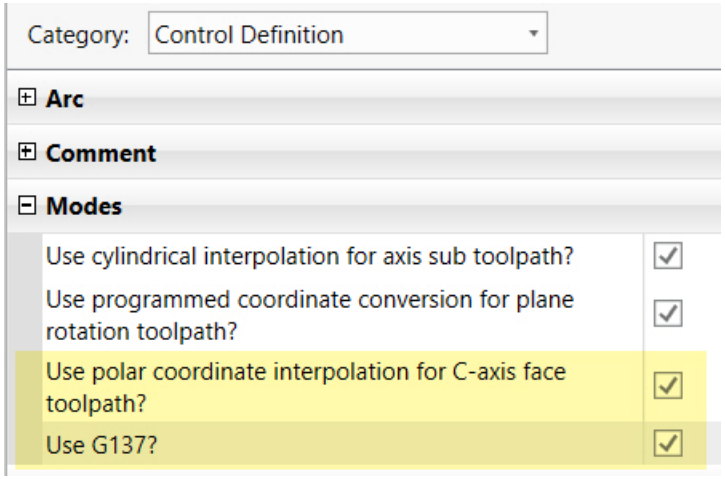
Note that your P300 **.machine** file includes an option for using polar interpolation for C-axis face drilling. This option should not be used and can be ignored.

You can also enable or disable these cycles by default. Follow these steps:

1. Open the **.machine** file in Code Expert.
2. Double-click the **Consumer** layer.



3. Select **Category: Control Definition**.
4. Go to the **Modes** section.
5. Select the **Use polar coordinate interpolation…** option to output `G101/G102/G103` as the default mode for C-axis face operations.
6. Press [**Ctrl+S**] to save the **.machine** file.

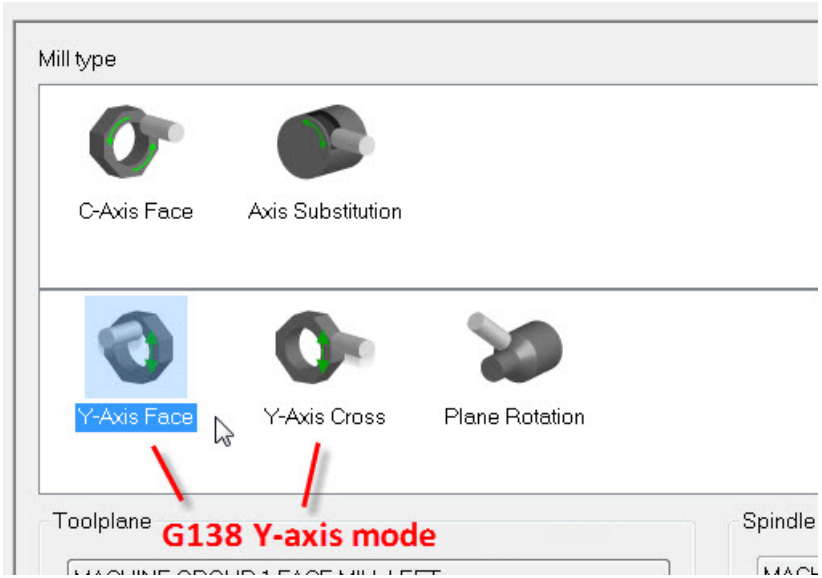


Y-axis (G138) mode

Your **.machine** file supports G138 mode for Y-axis face and cross operations.

This is automatically enabled when you select **Y-Axis Face** on **Y-axis Cross** on the **Setup** page for milling operations.

Unlike the other mill cycles, there are no Sync Manager options that let you configure the output in the Sync Manager.



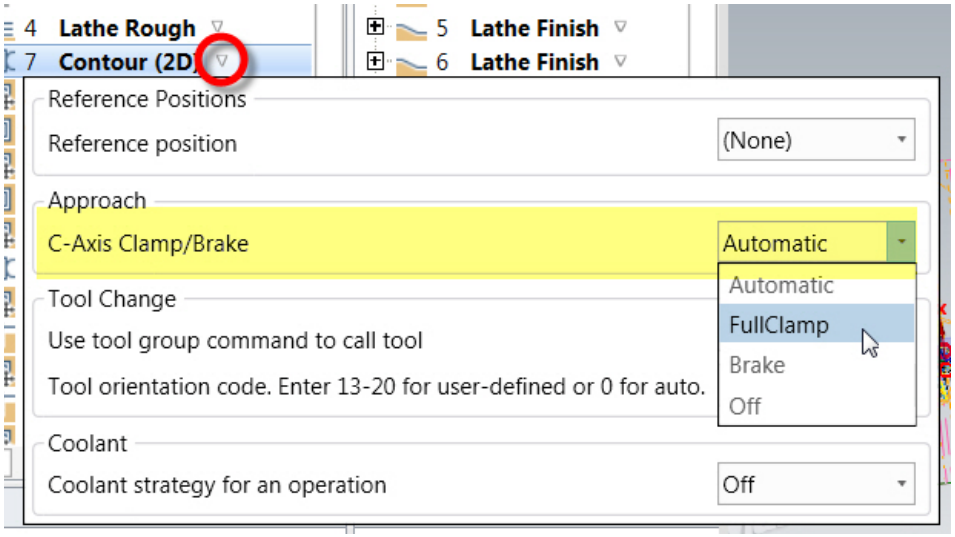
Home

C-axis clamping (M147)

Your **.machine** file lets you control the C-axis clamping mode (M147) for milling operations.

Set this option in the Sync Manager. Click the small triangle next to the toolpath name and select the desired **C-axis Clamp/Brake** mode:

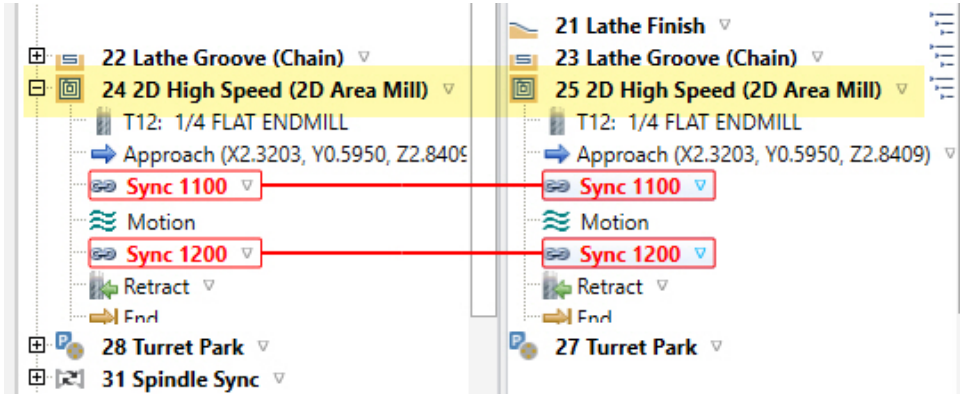
- **Automatic**—Mastercam will decide whether to clamp the C-axis based on the application and toolpath type.
- **FullClamp**—Force M147 output.
- **Off**—Suppress M147 output for the selected operation.
- **Brake**—This is not supported for the P300 **.machine** file.



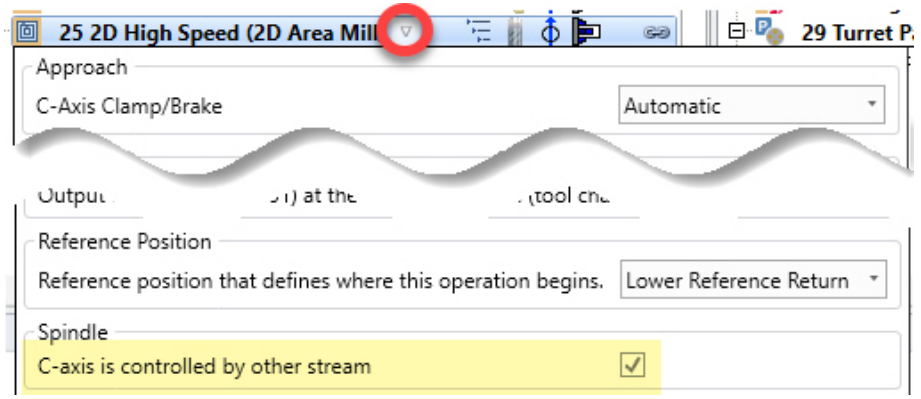
Home

Spindle commands for pinch/balanced milling operations

When you are machining with both turrets on the same spindle in a synchronized manner, you need to designate which stream controls the C axis. In this example, the upper and lower streams are machining on the main spindle at the same time.



Locate the operation in the dependent stream or turret—in this example, operation 25—and select the **C-axis is controlled by other stream** option.



In the dependent stream, your post outputs an `M153` interlock release command. This allows the main or sub spindle to rotate while the tool spindle is rotating. The other stream contains an `M110` command to turn on milling mode.

```
570  (Operation # 24)
571  (RD FACE ULT)
572  N12 (12  |  1/4 FLAT ENDMILL  | DIA. -  0.25)
573  NP007
574  M110
575  G00 X=VPVLX (X AXIS RETURN HOME)
576  G00 Z=VPVLZ (Z AXIS RETURN HOME)
577  NA12 T121212
578  SB=5000 M13
579  G00 X=VPVLX (X AXIS RETURN HOME)
580  G00 Z=VPVLZ (Z AXIS RETURN HOME)
581  C330. M16
582  M147
583  G138
584  G94 G19 Z-2.3203
585  X2.8409 Y.595
586  P1100
587  X.9409
588  G01 X.7409 F40.
589  X.7353 Y.5948
590  X.7297 Y.5944
```

```
551  (Operation # 25)
552  (RD FACE LLT)
553  N12 (12  |  1/4 FLAT ENDMILL  | DIA. -  0.25)
554  NP014
555  M153
556  G00 X=VPVLX (X AXIS RETURN HOME)
557  G00 Z=VPVLZ (Z AXIS RETURN HOME)
558  NB12 T121212
559  SB=5000 M13
560  G00 X=VPVLX (X AXIS RETURN HOME)
561  G00 Z=VPVLZ (Z AXIS RETURN HOME)
562  G138
563  G94 G19 Z-2.3203
564  X2.8409 Y.595


565  P1100
566  X.9409
567  G01 X.7409 F40.
568  X.7353 Y.5948
569  X.7297 Y.5944
```

While `M153` is active in the dependent stream, the following commands will result in an error:

- `M109` (turning mode), `M110` (milling mode)
- `M147`/`M146` (C-axis clamp/unclamp)
- Any C-axis move

Home

C-axis rotation direction

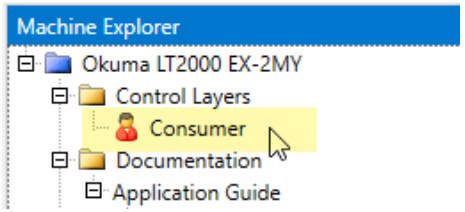
Your Okuma **.machine** file lets you specify how you want to rotate the C-axis:

- `M15/M16` (CW/CCW) commands
- `M960` shortest distance mode

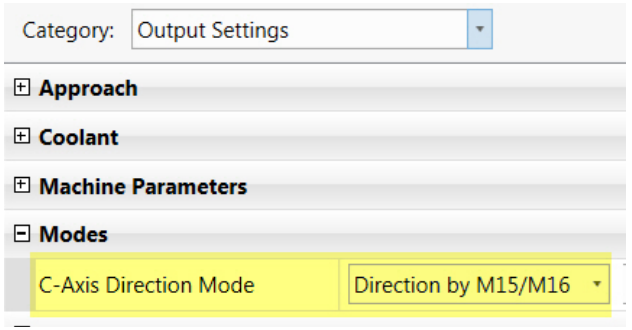
The `M960` feature is supported in the P300 control. If your machine does not support `M960`, select the `M15/M16` option.

Follow these steps:

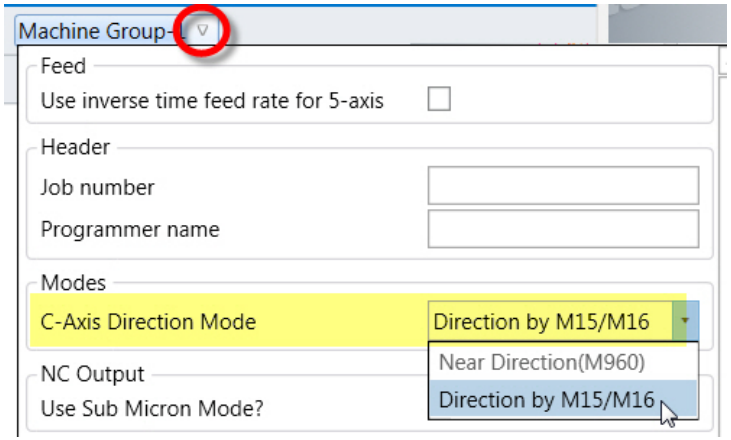
1. Open the **.machine** file in Code Expert.
2. Double-click the **Consumer** layer.



3. Select **Category: Output Settings**.
4. Go to the **Modes** section.
5. Select the desired **C-axis Direction Mode**.
6. Press [**Ctrl+S**] to save the **.machine** file.



You can also override the default setting for your part. Click the small triangle next to the machine group name and select the desired **C-axis Direction Mode**.



H: Configuring multiaxis operations

Your **.machine** file includes a number of options that you can use to configure how your post handles multiaxis toolpaths. You can edit the default settings in the **.machine** file, and you can also set them individually for each operation.

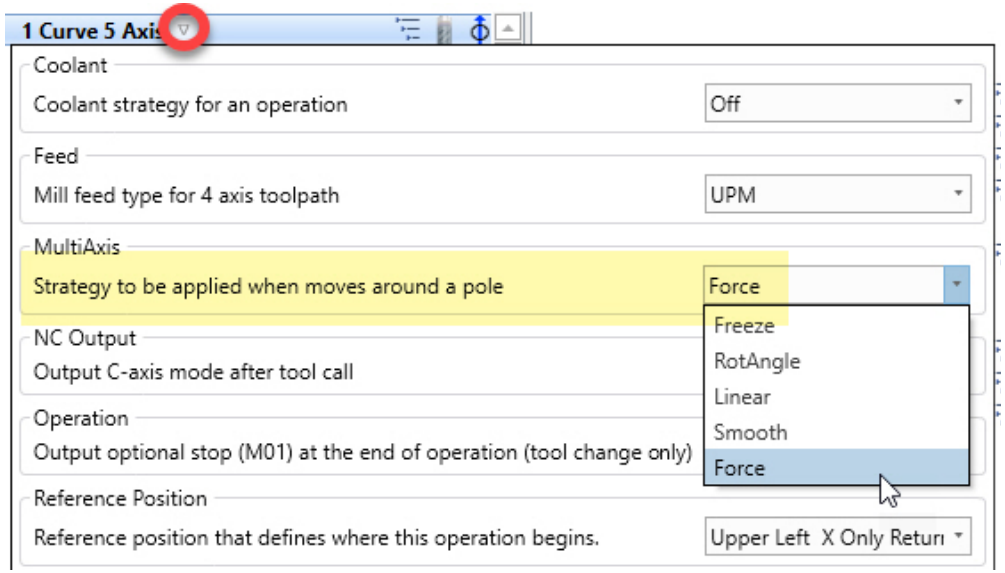
Full multiaxis output is typically only possible with machines that have a B-axis head. However, you can create multiaxis toolpaths with turret-only machines if you select 3-axis or 4-axis output.

- **Pole handling**
- **Feed rates for multiaxis operations**

Pole handling

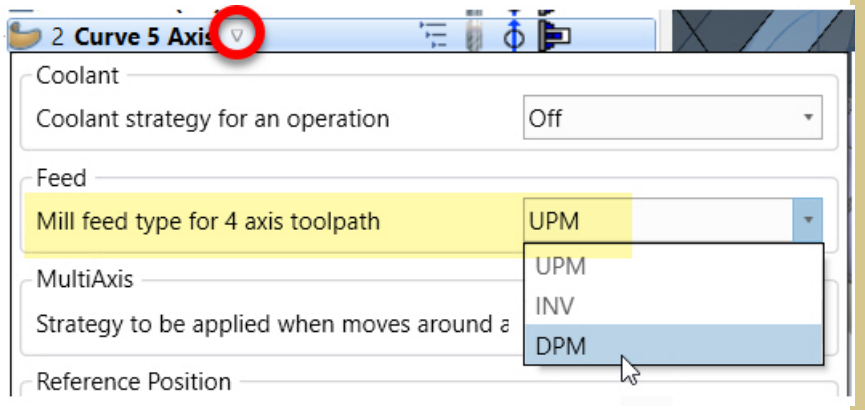
A pole (or "singularity") occurs when a machine has a C-axis that is rotating around Z, and the tool orientation is almost parallel to the Z-axis.

Pole handling should handle those cases when the tool axis and the spindle axis are colinear. That means that both directions are parallel and a mathematical singularity is reached. For turret-only machines, select the **Force** option to output proper X/Y and C coordinates.



Feed rates for multiaxis operations

You can configure the feed rate units for each multiaxis operation. Click the small triangle next to the operation name and select either **UPM** (units per minute—either inch or mm) or **DPM** (degree per minute). **INV** (inverse feed rate) is not supported on turret-only machines and should not be selected.



Home

I: Configuring turning operations

The following sections describe Okuma options and modes that let you configure turning operations.

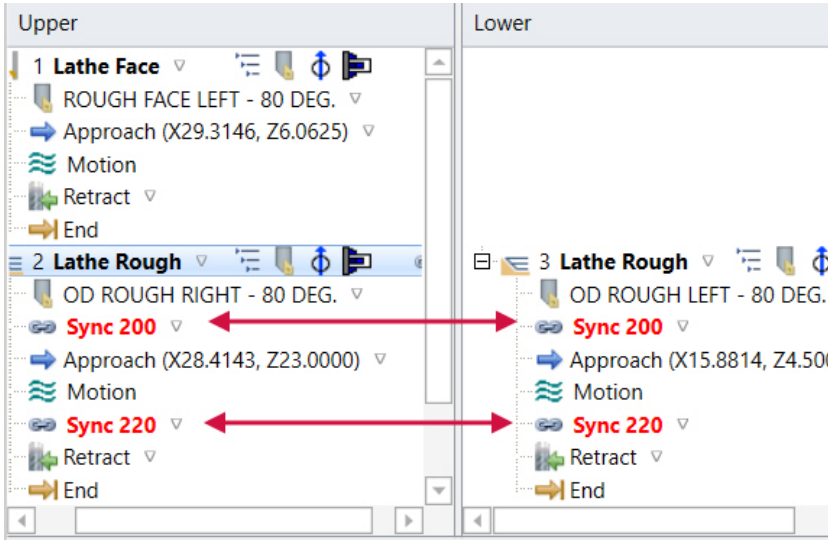
- Turning with both turrets on the same spindle (non-pinch turn)
- Entering a Y offset
- Turning on the tool spindle for lathe drill operations
- Infeed pattern for lathe threading

Home

Turning with both turrets on the same spindle (non-pinch turn)

When both turrets are working on the same spindle at the same time, follow this outline so that your post can correctly process each stream.

1. Make sure that the spindle speed is the same for both operations.
2. Create syncs between the **Approach** and **Retract** nodes of each operation. Sync the operations carefully to ensure that you do not attach the syncs to the start/end of the operation or other node.
3. Press [**Ctrl+S**] to save the syncs back to your part file.



Mastercam syncs the operations as shown below. In addition to the syncs that you created in Step 2, Mastercam also outputs syncs or wait codes with the spindle speed commands.

```
579  P1000
580  SYNC=12
581  G00 X=VPVLX (X AXIS RETURN HOME)
582  G00 Z=VPVLZ (Z AXIS RETURN HOME)
583  NA16 T161616
584  G18
585  G97 S3000 M03 M42 P1005
586  G95 G00 Z-1.1875
587  X1.64
```

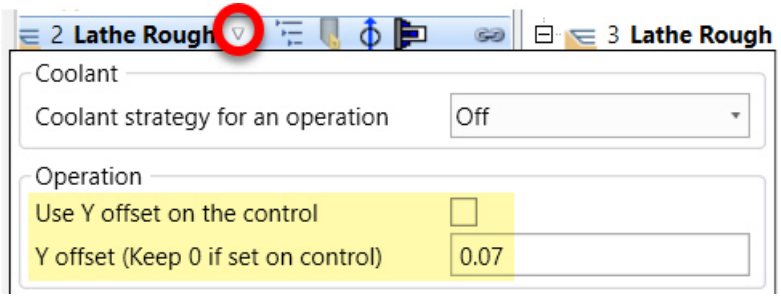
```
504  (Operation # 24)
505  N16 (16  |  OD GROOVE RIGHT - NARROW  |  GC-4125)
506  NP014
507  P1000
508  SYNC=12
509  G00 X=VPVLX (X AXIS RETURN HOME)
510  G00 Z=VPVLZ (Z AXIS RETURN HOME)
511  NB16 T161616
512  G18
513  G97 S3000 M03 M42 P1005
514  G95 G00 Z-3.1875
515  X2.14
```

```
2434  P1000
2435  SYNC=3
2436  M110
2437  G00 X=VPVLX (X ONLY RETURN)
2438  NB14 T141414
2439  SB=6000 M13
2440  G00 X=VPVLX (X ONLY RETURN)
2441  G94 G00 C227.9033 M16
2442  G17 Z2.
```

Entering a Y offset

Your **.machine** file lets you enter a Y-axis offset for turning operations. Follow these steps:

1. In the Sync Manager, click the small triangle next to any turning operation.
2. Deselect the **Use Y offset on the control** option.
3. Enter a **Y offset** value.
4. Press [**Ctrl+S**] to save the settings back to your part file.

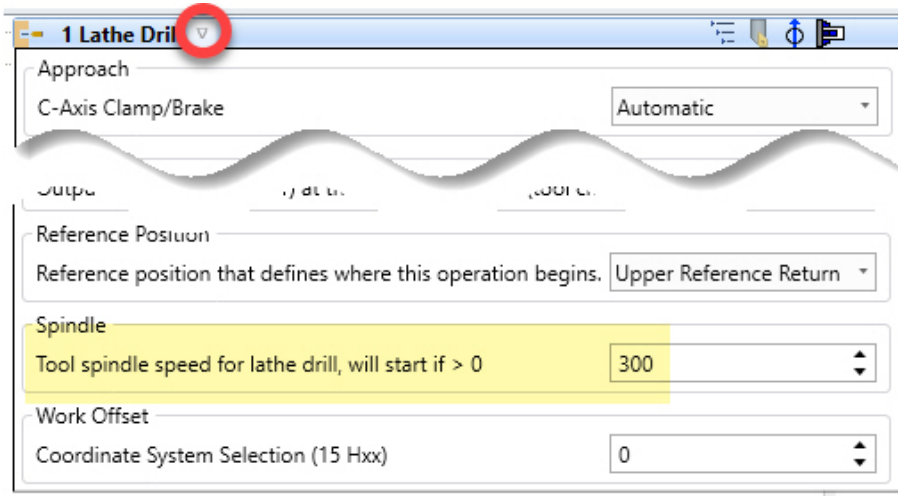


Home

Turning on the tool spindle for lathe drill operations

For machines that are equipped for live tooling, your **.machine** file gives you the option of turning on the tool spindle for lathe drilling operations. Follow these steps:

1. In the Sync Manager, click the small triangle next to any turning operation.
2. Enter the **Tool spindle speed for lathe drill**…. Entering any value greater than zero will turn on the tool spindle at the selected RPM.
3. Enter a **Y offset** value.
4. Press [**Ctrl+S**] to save the settings back to your part file.



Your post automatically outputs an `M153` command to turn off the tool spindle interlock, enabling the spindle to be turned on (`M13`). After the spindle is turned off the spindle interlock is turned back on.

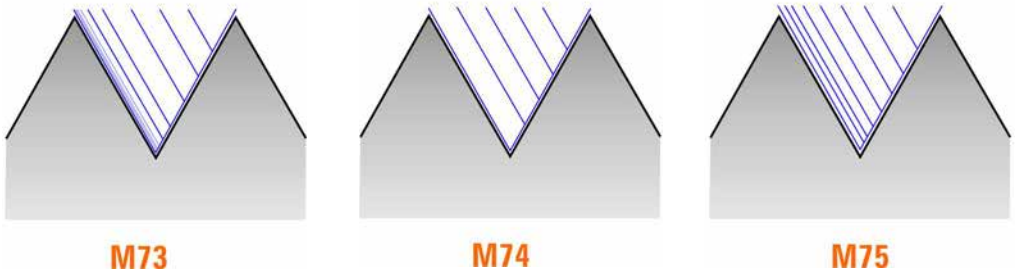
```
G13
M331
G140
(Operation # 1)
N1 (1   |  SPOT TOOL  .25 DIA.)
NP001
M153
G00 X=VPVLX (X AXIS RETURN HOME)
G00 Z=VPVLZ (Z AXIS RETURN HOME)
NA01 T10101
G18
G97 S200 M03 M41
M13 SB=300
G95 G00 Z.25
X0.
G74 X0. Z-1. K.15 D.125 F.01
M05
M12
M152
T0100
```

Home

Infeed pattern for lathe threading

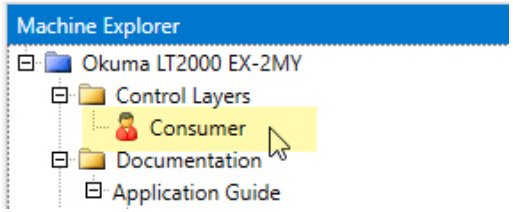
Your Okuma **.machine** file lets you choose the infeed pattern for lathe threading:

- `M73`—Constant cut depth until near end, then successive passes at 1/2, 1/4, and 1/8 depth.
- `M74`—Constant cut depth.
- `M75`—Decreasing cut depths to maintain optimum chip load.

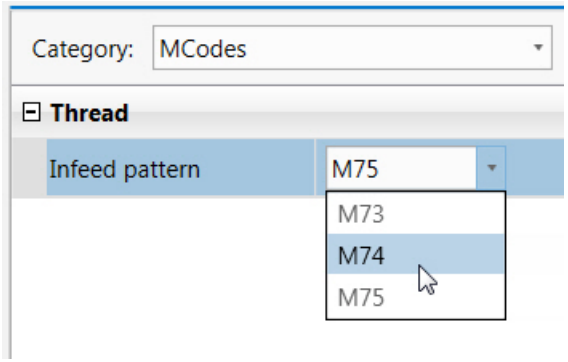


Follow these steps:

1. Open the P300 **.machine** file in Code Expert.
2. Double-click the **Consumer** layer.



3. Select **Category: Output Settings**.
4. Go to the **Modes** section.
5. Select the desired **C-axis Direction Mode**.
6. Press [**Ctrl+S**] to save the **.machine** file.

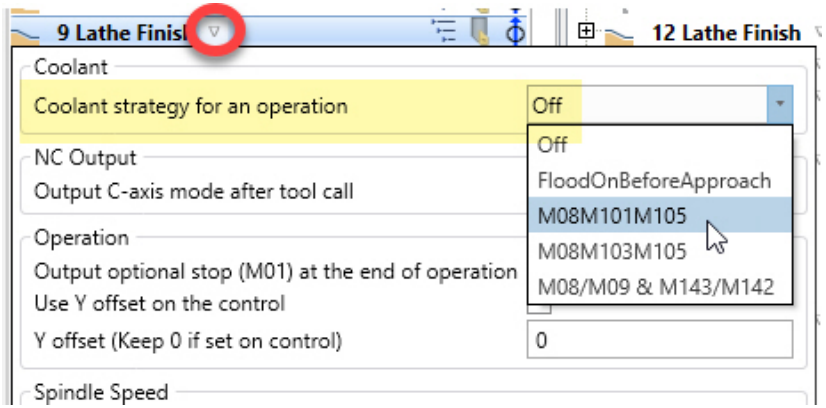


J: Using coolant

If you are familiar with other Mastercam products, you are used to selecting coolant options inside Mastercam as part of the toolpath parameters. In Mill-Turn, coolant selection is done in the Sync Manager; the **Coolant** button is not present inside Mastercam.

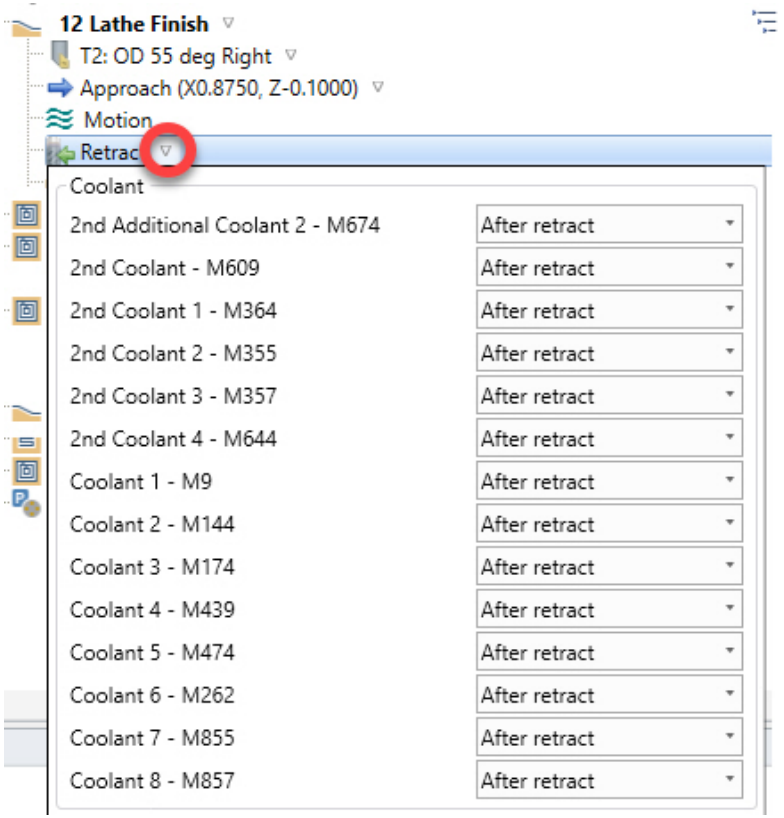
Most Mill-Turn coolant options are organized into *strategies*. For each operation, click the small triangle next to the operation name and select the desired strategy. Your post automatically outputs the proper coolant off codes at the end of the operation. Select Off only when you want to force output of the coolant off commands.

Note that the coolant options that you see for your machine might be different than the ones shown in this picture.



Additional coolant options

In addition to the coolant strategies, your **.machine** file supports a number of other coolant options. These are available by clicking on the **Retract** node for each operation.

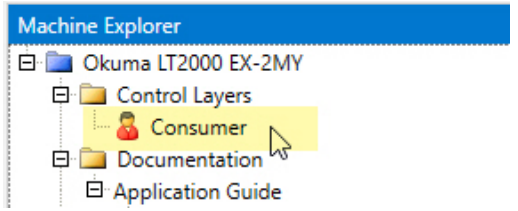


Home

Default coolant options

You can select a default coolant strategy and save it in the **.machine** file. Note that the coolant options that you see for your machine might be different than the ones shown in this picture.

1. Open the **.machine** file in Code Expert.
2. Double-click the **Consumer** layer.



3. Select **Category: Output Settings**.
4. Go to the **Coolant** section.
5. Set the desired default strategy.
6. Press [**Ctrl+S**] to save the **.machine** file.

